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Auctioning the Digital Dividend: A model for spectrum auctions

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ABSTRACT

We model a spectrum auction where firms purchase units to participate in a constrained, multi-product, downstream market. We use dynamic programming techniques to numerically solve for the optimal bidding strategy in a clock auction. Firms value constraining competitor market power, so incumbents often bid aggressively to shut out entrants. We find that high cost firms may hold up the market, so the auction may be inefficient and generate zero revenue. An auction may be optimal for a regulator maximising total surplus. A regulator maximising auction revenue sets reserve prices high enough to restrict spectra sold, effectively behaving as a monopolist.

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1. Introduction

In most modern economies, the "Digital Dividend" represents a substantial technological and financial windfall. Developments in television broadcasting allow the

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transmission of digital video and sound that require only one-sixth the bandwidth of inferior quality analog transmissions. Therefore, replacing analog transmissions with digital transmissions frees up a large quantity of spectrum frequencies, which can be employed to transmit wireless data, used by mobile phones, laptop computers, and other devices.

These unallocated new spectra offer opportunities for consumers, producers, and government. First, wireless telecommunications companies can expand their services. Firms that have access to the new spectra can effectively provide superior data services. Second, consumers can enjoy a wider range of services. Third, the government benefits from the unallocated spectra in two ways. On the one hand, the government, as an auctioneer, can earn revenue from the auction. On the other hand, the government, as a regulator, can affect the degree of competition in the telecommunications market to increase total surplus. A common policy tool adopted by governments to achieve social optimum is to conduct an auction with a reserve price and caps on firms' winnings to prevent one firm from winning all units and becoming a monopolist; see Cramton et al. (2011).

Spectrum auctions are potentially an efficient way to allocate the new spectra across firms and do not represent new phenomena in most countries.[1] A spectrum auction is an example of a *multi-unit auction*. The auctioneer is selling a collection of relatively homogeneous goods to multiple firms.[2] Our aim in this paper is to examine the equilibrium properties of a Digital Dividend auction where participants compete for market power in the telecommunications market. Our model consists of a downstream market, in which firms play a Cournot game with capacity constraints, similar to Laye and Laye (2008). Specifically, firms produce two goods: low and high data use plans, and are constrained in their ability to produce data plans by the amount of spectra they have available. Having projected their potential profits in the downstream market, firms enter into a simultaneous uniform-price clock auction to increase their production capacity.

It is worth noting that although we consider the spectrum auction as the motivating example, our model can be generalised to investigate other problems with an interaction between a downstream market and an upstream auction/competition. Examples include procurement auctions (to obtain market access) for medical drugs in third-world countries and competition by airlines for landing slots; see Eső et al. (2010).

Given the complexity of solving multi-unit auctions (see Cramton, 2013), we first investigate a special case where only 2 firms compete for 2 units. For our subsequent results, we use a numerical technique to solve for the bidders' optimal strategies, and the resulting allocations.

[1] The United States began its auctions for spectrum licences during the 1990s, and have since assessed their efficiency. For example, see Cramton (1997), Cramton (1998), Cramton (2013), Kwerel and Rosston (2000), Bush (2010) and Cramton et al. (2011). Meanwhile, the GSM (second generation mobile telecommunication) and UMTS (third generation) auctions in Europe from 1999 to 2001 attracted a lot of attention from the public for their interesting outcomes. For example, see van Damme (2002), Klemperer (2005), and Grimm et al. (2003). Economists have also been surprised by the huge revenues the British government realised from the sale of its 3G telecom licences (Binmore and Klemperer, 2002).

[2] In reality, spectrum frequencies are not truly homogeneous because technology-dependent synergies from having access to adjacent frequencies exist. In addition, regional standards, technological limitations, and device manufacturers' decisions may make certain segments of a given band more or less desirable.

We find many instances of firms following mixed strategies in their bidding. In these cases, our model generates *distributions* of allocations, profits, levels of social welfare, and revenue for a regulator/auctioneer (henceforth regulator). We consider several scenarios for market structure and investigate the equilibrium outcomes using these measures.

We find that when there is only one high cost firm in the market, the low cost firms can successfully obtain market power by winning enough spectrum units. More importantly, when there are multiple high cost firms present, they can hold up the market by threatening to raise prices (which is harmful to low cost firms who seek to win larger numbers of units). These results illustrate why some recent spectrum auctions have resulted in socially suboptimal allocations of spectra: downstream market structure matters for auction efficiency.

Next, we consider a case where a regulator cannot observe the type of firms due to informational asymmetries. Our findings show that while caps are useful to account for the degree of competition in the downstream market, a reserve price is often set to increase revenue. Therefore, a regulator concerned with auction revenue would set the reserve price high and would limit production, effectively acting like a monopolist.

Our work contributes to the literature in several ways. First, potential market power in the downstream market may be of concern to a benevolent regulator who also happens to be the auctioneer. Similar to the literature on licence auctions, a relationship exists between auction outcomes and market prices.[3] Offerman and Potters (2000) illustrate this relationship using experiments and conclude that bidders' valuations depend not only on their licenced units but also on whether the remaining units are obtained by someone else and by whom they will be won. Similar to their work, we incorporate the downstream market into our auction model. However, in our case (as discussed in Section 2), we examine spectrum units (which constrain production) rather than licences (which allow participation in the market).

Second, at multi-unit auctions, bidders are often assumed to have non-increasing marginal valuation (NIMV) for the goods in question.[4] This is a classic assumption in economic theory to make models tractable. This assumption is not necessarily valid in our case because marginal value of a spectrum unit may increase with market power. In our model, units sold in the auction affect the production capacity. Therefore, winning more units not only increases a firm's production capacity, but also limits its competitors' capacities, which gives it market power. In such a case, the marginal value of an additional spectrum unit may *increase* as a firm gets closer to being a monopoly. This effect of market power on marginal valuations is also analysed by Eső et al. (2010). Assuming complete information, they model a downstream industry where firms compete to buy capacity in an upstream Vickrey–Clarke–Groves auction. Our approach is similar to theirs, but we allow firms to compete in a multi-product industry, and adopt the

[3] For example, see Janssen and Karamychev (2007), Hoppe et al. (2006), Janssen (2006), and Jehiel and Moldovanu (2003).

[4] For example, see Kastl (2011), Hortaçsu and McAdams (2010), Mishra and Parkes (2009), Blume et al. (2009), Riedel and Wolfstetter (2006), and Katzman (1999).

multi-unit clock auction as the mechanism for spectrum allocation, which is more applicable in practice. Similar to Eső et al. (2010), we find that an auction may lead to aggressive bidding, which may result in constrained firms, and a decrease in downstream market total surplus, relative to assigning spectra.

Finally, we assume uncertainty regarding firms' costs, on the part of a regulator. We examine how the regulator's objective function affects the optimal reserve price and cap structure in the auction. A trade-off may exist between the total surplus and auction revenue. In particular, even if an auction is efficient, it may result in an inefficient downstream market, where one firm wins most of the units. Therefore, the regulator aims to maximise some combination of total surplus in the downstream market and auction revenue. Dana and Spier (1994) focus on a risk-neutral regulator's problem of choosing who produces in a downstream market. We consider a more general objective function, and allow the regulator to be risk-averse.

The layout of the remainder of this paper is as follows. Section 2 outlines our model for the downstream market and the auction itself. Section 3 presents a 2-firm, 2-unit example. Section 4 outlines our numerical technique for solving auctions in general. Section 5 contains results from analysing uncapped auctions with no reserve price, while Section 6 explores the role of the auctioneer. Lastly, Section 7 concludes.

2. Downstream market

We begin by describing the telecommunications market, which we will refer to as the *downstream market* throughout this paper. Suppose M firms operate in the market, which is composed of two products: high and low data use plans (henceforth, products). Let q_{hi} and q_{li} denote firm i's output for the high and low products. Cost of production may differ across firms as well as products. Thus, let $C_{hi}(q)$ and $C_{li}(q)$ represent firm i's total cost of producing q units of the respective products. $c_{hi}(q)$ and $c_{li}(q)$ denote firm i's marginal costs.

The demand functions for each of the two products depends only on the price of that product. Hence inverse demand for the two products is described as follows:

$$\text{Low Product: } P_l = P_l(\mathbf{q_l}; \boldsymbol{\beta}_l),\ \mathbf{q_l} = (q_{l1}, q_{l2}, \ldots, q_{lM}) \tag{1}$$

$$\text{High Product: } P_h = P_h(\mathbf{q_h}; \boldsymbol{\beta}_h),\ \mathbf{q_h} = (q_{h1}, q_{h2}, \ldots, q_{hM}). \tag{2}$$

In Eqs. (1) and (2), $\mathbf{q_l}$ and $\mathbf{q_h}$ are the vector of outputs for each product by all firms, whereas $\boldsymbol{\beta}_l$ and $\boldsymbol{\beta}_h$ are the corresponding demand parameters. Note that if all firms' outputs for a particular product are perfect substitutes, then one can simplify the inverse demand functions so that price depends only on the aggregate production.[5]

[5] Alternatively, the demand curves themselves could be generated by considering a continuum of customers who switch between the products as prices rise/fall. In addition, one could argue that the product prices may depend on both vectors of outputs, as the two products may be complements or substitutes.

Spectrum *licences* are often seen as complements (due to their geographical position). However, spectrum *units* (in a given location) are substitutes. Since the Digital Dividend opens up large quantities of spectra, auctions are often held for multiple units in the same location.[6] In this paper, we treat spectrum units as substitutes. Firms in our model compete in the auction not only to gain access to the market, but also to increase their production capacity.

We assume that firms are endowed with an initial allocation of legacy spectra, which we denote as $\mathbf{L} = (L_1, L_2, \ldots, L_M)$. We further assume that firms' initial endowments are already paid for, or can be regarded as sunk costs. Meanwhile, depending on the auction outcome, firms may win new spectrum units ($\mathbf{B} = (B_1, B_2, \ldots, B_M)$). Marginal productivity is 1 for both the new and legacy spectra. Hence, using the legacy and new spectra, firm i faces the following capacity constraints:

$$q_{hi} \leq B_i \tag{3a}$$

$$q_{hi} + q_{li} \leq B_i + L_i \tag{3b}$$

$$q_{hi}, q_{li} \geq 0. \tag{3c}$$

Firm i can use new spectra to increase production of the high product, the low product, or both, whereas legacy spectra can only be used in the low product market.[7] Finally, the last two inequalities ensure non-negative production for the two products.

Given the spectrum allocations ($\mathbf{B}$, $\mathbf{L}$), we can write down firm i's profit in the downstream market as follows:

$$\Pi_i^D(q_{hi}, q_{li}) = P_h(\mathbf{q_h}; \boldsymbol{\beta}_h)\ q_{hi} - C_{hi}(q_{hi}) + P_l(\mathbf{q_l}; \boldsymbol{\beta}_l)\ q_{li} - C_{li}(q_{li}) \tag{4a}$$

$$\Pi_i(q_{hi}, q_{li}) = \Pi_i^D(q_{hi}, q_{li}) - C_a(B_i, P_a) \tag{4b}$$

where both profit functions are defined conditional on the vector of other firms' supply ($\mathbf{q_{h,-i}}, \mathbf{q_{l,-i}}$), and the vector of legacy and new spectra allocated to all firms ($\mathbf{B}$, $\mathbf{L}$). Eq. (4a) is the firm's profit in the downstream market, whereas the firm's profit net of the cost of new spectra is given in Eq. (4b). The right-hand side of Eq. (4a) is the sum of the profits from the high and low products, and the last term in Eq. (4b) is the cost of new spectra, which depends on the auction price P_a.

We assume a complete information environment for the auction as well as the downstream market. All the information regarding the cost of production, the spectrum allocations, and the demand parameters, are known by all firms. Therefore, firms can first solve for their profits in the downstream market, given the spectrum allocations. Then, they

[6] In small countries, there may indeed be only one location. See Cramton (2013) for further discussion of the complement/substitute nature of spectra.

[7] The model could be embellished to allow for more complex substitutions between spectrum units. For example, it is straightforward to put a weight on the B_i terms, and another weight on the L_i term in Eq. (3), so that different spectrum units have different marginal productivity.

can enter the auction for the new spectra using information on downstream valuations to formulate bidding strategies.

2.1. Profit maximisation in downstream market

Given the spectrum allocations, we assume that the firms compete in a Cournot game by deciding how much to produce for both high and low products. Because firm outputs for the high and low products are limited by the capacity constraints described in Eq. (3), the distribution of the new and legacy spectra across firms can result in market power. First, we describe the feasible set for firm i:

$$\mathcal{A}_i = \{(q_{hi}, q_{li}) : \ (q_{hi}, q_{li}) \text{ satisfy the constraints in (3) given } (\mathbf{B}, \mathbf{L})\}$$

Using this feasible set, firm i's profit maximisation problem is as follows:

$$\max_{(q_{hi}, q_{li}) \in \mathcal{A}_i} \ \Pi_i^D(q_{hi}, q_{li} | \mathbf{q_{h,-i}}, \mathbf{q_{l,-i}}, B_i, \mathbf{B_{-i}}, \mathbf{L}) \tag{5}$$

where firm i chooses the outputs $(q_{hi},\ q_{li})$ from the feasible set $\mathcal{A}_i$, conditional on the spectrum allocations. In this way, we can derive the payoffs for firms, which we then use to solve for the auction equilibrium. We first define the equilibrium in the downstream market.

Definition 2.1. A Nash Equilibrium in pure strategies in the Cournot game for the downstream market is such that given (**B, L**):

$$\{q_{hi}^*, q_{li}^*\}_{i=1}^M \text{ solves (5)}; \forall \ i = 1, \ldots, M.$$

Given the equilibrium behavior of all other firms $\left(\mathbf{q}^*_{\mathbf{h},-\mathbf{i}}, \mathbf{q}^*_{\mathbf{l},-\mathbf{i}}\right)$, firm i does not find it profitable to change its production decision. Furthermore, if we assume that demand and cost functions are differentiable, then we can solve for the Nash equilibrium using the following complementarity problem:

$$\frac{\partial \Pi_i^D(q_{hi}, q_{li}, \cdot)}{\partial q_{hi}} + \eta_i - (\lambda_i + \mu_i) = 0; \forall i = 1, \ldots, M \tag{6a}$$

$$\frac{\partial \Pi_i^D(q_{hi}, q_{li}, \cdot)}{\partial q_{li}} + \xi_i - \lambda_i = 0; \forall i = 1, \ldots, M \tag{6b}$$

$$B_i - q_{hi} \geq 0 \perp \mu_i \geq 0; \forall i = 1, \ldots, M \tag{6c}$$

$$B_i + L_i - (q_{hi} + q_{li}) \geq 0 \perp \lambda_i \geq 0; \forall i = 1, \ldots, M \tag{6d}$$

$$q_{hi} \geq 0 \perp \eta_i \geq 0; \forall i = 1, \ldots, M \tag{6e}$$

$$q_{li} \geq 0 \perp \xi_i \geq 0; \forall i = 1, \ldots, M \tag{6f}$$

where $\{\mu_i, \lambda_i, \eta_i, \xi_i\}$ are the Lagrange multipliers on the capacity constraints given in Eq. (3). The complementarity problem (6) is a square problem with $6 \times M$ equations and $6 \times M$ unknowns $\{q_{hi}, q_{li}, \mu_i, \lambda_i, \eta_i, \xi_i\}_{i=1}^{M}$.

As shown by Monderer and Shapley (1996), the equilibrium in these potential games may not be unique. However, Laye and Laye (2008) demonstrate the uniqueness of the Cournot–Nash equilibrium in a multi-product market with linear demands, positive convex production costs, and closed convex sets of capacity constraints. For the remainder of the paper, we assume linear cost and demand functions, which lead to a unique equilibrium in the downstream market.

Before we move on to the details of the auction, it is worth noting that bidders' valuations of each spectrum unit not only depend on each others' winnings (since the firms will compete in the downstream market), but may also violate the non-increasing marginal valuation assumption. For example, consider equilibria in which firm i is unconstrained ($\mu_i = \lambda_i = 0$), and all other firms are constrained in their production of the high product ($\mu_k > 0$), but not the low product ($\lambda_k = 0$). Therefore, firm k's production will satisfy $q_{hk} = B_k$. Suppose that firm i winning an extra unit of spectrum will mean that firm j will win one unit less. Given the linear inverse demand for each product $\left(P_n = a_n - b_n \sum_{k=1}^{M} q_{nk}, n = h, l\right)$, and constant marginal costs $(c_{ni}, n = h, l)$, firm i's profit in the downstream market (4a), conditional on the other firms' actions, simplifies to:

$$\Pi_i^D = q_{hi}\left(a_h - b_h \sum_{k \neq i} B_k - b_h q_{hi} - c_{hi}\right) + q_{li}\left(a_l - b_l \sum_{k \neq i} q_{lk}^* - b_l q_{li} - c_{li}\right).$$

We can then calculate firm i's marginal profit (6a) and best response:

$$\frac{\partial \Pi_i^D}{\partial q_{hi}} = a_h - b_h \sum_{k \neq i} B_k - 2b_h q_{hi} - c_{hi} = 0$$

$$\Rightarrow \quad q_{hi}^* = \frac{1}{2b_h}\left(a_h - b_h \sum_{k \neq i} B_k - c_{hi}\right),$$

which yields $dq_{hi}^*/dB_j = -1/2$. Denote the equilibrium downstream profit for firm i as Π_i^{D*}. By the envelope theorem, we get $\frac{d\Pi_i^{D*}}{dB_j} = -b_h q_{hi}^* < 0$. Further, we can calculate the second derivative

$$\frac{d^2\Pi_i^{D*}}{dB_j^2} = \frac{\partial^2 \Pi_i^D}{\partial B_j^2} + 2\frac{\partial^2 \Pi_i^D}{\partial q_{hi}^* \partial B_j}\frac{dq_{hi}^*}{dB_j} + \frac{\partial^2 \Pi_i^D}{\partial q_{hi}^{*2}}\left(\frac{dq_{hi}^*}{dB_j}\right)^2 = \frac{b_h}{2} > 0.$$

Hence, with a marginal increase in B_i leading to a decrease in B_j, this implies that firm i has an increasing marginal valuation of B_i:

$$\frac{d\Pi_i^{D*}}{dB_i} > 0, \text{ and } \frac{d^2\Pi_i^{D*}}{dB_i^2} > 0.$$

2.2. Clock auction for the new spectra

We adopt a multi-unit English (ascending price, clock) auction in this paper.[8] Suppose that there are N items to be sold. The auctioneer's goal is to find a price such that there is no excess demand for the goods; see Cramton (2013). In general, the auctioneer may make use of a reserve price and caps. Spectrum caps are widely used in European 3G auctions and can be useful to help avoid monopoly formation (or market power) in the downstream market; see Cramton et al. (2011). Meanwhile, a reserve price can be set to ensure that allocated spectra are utilised (as opposed to zero price leading to incentives to limit others' production).

Let P_a represent the current price in the auction. An English auction works in the following way:

1. The auctioneer starts P_a at the reserve price ($\underline{P_a}$) and firms bid their caps ($\overline{\mathbf{B}}$).[9]
2. At the current price P_a, firms submit new bids, which cannot exceed their existing bids:

$$\mathbf{B}(P_a) = (B_1(P_a), \ldots, B_M(P_a))$$
$$B_i(P) \leq B_i(P'); \text{ for } P \geq P', \forall i = 1, 2, \ldots, M.$$

   Firms may respond to each others' bids at the ongoing price. How we model this is discussed in more detail in Section 4. We assume no two firms lower their demand at the same time. If two (or more) firms try to simultaneously lower demand, then they are each equally likely to lower demand first.
3. If there is excess demand at the current price, then the auctioneer increases the price:[10]

$$\sum_{i=1}^{M} B_i(P_a) > N \;\Rightarrow P_a \text{ increases to } P_a + \Delta P.$$

4. Steps 2 and 3 are repeated until the price reaches P_a^* such that:

$$P_a^* = \inf\left\{P_a \;:\; \sum_{i=1}^{M} B_i(P_a) \leq N\right\}.$$

[8] While simultaneous multiple-round ascending price auctions are more commonly used (e.g., in the US and European Union), the two auction formats yield very similar results, as long as the units are good substitutes; see Cramton et al. (2011) for a discussion. The auction model described in this paper is used in New Zealand and Sweden.

[9] Allowing firms to open the auction bidding below their caps results in a pre-auction game with many possible actions for each firm. Given that firms subsequently lower their demand, there is no disadvantage to initially bidding the full cap. However, in an auction where bidders may be forced to buy units (such as a situation where bidding closes as soon as demand equals supply), this would not be the case, and bidders might bid below their caps initially.

[10] In US spectrum auctions, bids are increased in discrete amounts, generally 5–20% per increment. From one to twenty increases are conducted per day. During early rounds, when there is more information to respond to, fewer increases are used per day. During later stages, less time is allocated per increase (see Cramton, 2002).

When the auction ends, the equilibrium vector of spectrum allocation is $\mathbf{B^*}$. Firm i pays the auction price P_a^* for each unit it wins. Thus, the total cost of spectra to firm i is $C_a(B_i^*, P_a^*) = P_a^*\ B_i^*$. The total revenue collected by the auctioneer equals $P_a^*\ \sum_{i=1}^{M} B_i^*$, which may be less than $P_a^*\ N$, if the auction ends with an excess supply.

3. Example: auctioning 2 units with 2 firms

To provide intuition regarding our approach to the auction problem, we consider an environment where two firms are competing for two units. In this case (for simplicity), each firm produces only one product. Each unit of spectrum won in the auction allows a firm to produce up to one unit of the product. The market demand for the product is $P = 4 - q_1 - q_2$, where P denotes the price, and $\{q_1,\ q_2\}$ indicate firm outputs. Both firms have marginal cost $c_i = 1$.

Given this setup, a firm that wins both units will earn monopoly profits in the downstream market of 9/4. A firm that wins only one unit (while the other unit goes unsold) earns monopoly profits of 2. Finally, if both firms win one unit, they will each earn duopoly profits of 1.

With two spectrum units, there are several market structures that could eventuate: (1) either firm is a monopolist with 2 units, (2) either firm is a monopolist with 1 unit, or (3) both firms are duopolists (with 1 unit each). Table 1 indicates the respective payoff to each firm depending on the auction outcome and costs of production.

We investigate strategies by first considering low bid vectors, and then, using the equilibria for these, inferring how the firms will behave at high bid vectors. We solve this auction in three steps:

1. Given the current auction price P_a, we analyse how firms behave in a static game, *where the auction will clear regardless of their actions*. In this case, we examine the behaviour when $\mathbf{B} = (1, 1)$.

Table 1
Downstream market profits as a function of firm costs and auction winnings. Case 1 pertains to the example in Section 3, while Cases 2 and 3 are discussed in Appendix A.

Outcome	Case 1[a]		Case 2[a]		Case 3[a]	
	$c_1 = 1,\ c_2 = 1$		$c_1 = 10/9,\ c_2 = 10/9$		$c_1 = 1,\ c_2 = 10/9$	
	Π_1^D	Π_2^D	Π_1^D	Π_2^D	Π_1^D	Π_2^D
Monopoly (Firm 1 with 2 units)	9/4	0	$(13/9)^2$	0	9/4	0
Monopoly (Firm 2 with 2 units)	0	9/4	0	$(13/9)^2$	0	$(13/9)^2$
Monopoly (Firm 1 with 1 unit)	2	0	17/9	0	2	0
Monopoly (Firm 2 with 1 unit)	0	2	0	17/9	0	17/9
Duopoly	1	1	$(26/27)^2$	$(26/27)^2$	19/18	$(17/18)^2$

[a] The terms Π_1^D and Π_2^D denote the payoff of Firms 1 and 2, respectively.

Table 2
The finite state game when both firms are low cost producers ($c_1 = c_2 = 1$) and both firms are bidding one unit at auction price P_a.

		Firm 2	
		Drop	Not drop
Firm 1	Drop	$\frac{1}{2}\max(2-P_a,0)$, $\frac{1}{2}\max(2-P_a,0)$	0,$\max(2-P_a,0)$
	Not drop	$\max(2-P_a,0)$, 0	$1-P_a$, $1-P_a$

2. Next, we examine how the firms determine the price at which the auction will progress to (1,1). At bid (2,1), firm 1 decides at which price the (1,1) game will be played. If, instead, the bid vector is (1,2), firm 2 makes this decision.
3. Given this behaviour in the (1, 2) and (2, 1) games, we determine how firms react in the (2, 2) game.

3.1. The (1,1) game

In this game, each firm currently demands one unit in the auction. Since total demand equals supply (2 units), the auction will clear immediately if neither firm drops. If one firm drops, the other can respond by dropping, or allow the auction to clear with it holding 1 unit.[11] Table 2 illustrates firm payoffs.

Given the auction price P_a, we can analyse the equilibria of this game in three regions:

1. If $P_a > 2$, the price is sufficiently high that each firm has a weakly dominant strategy to drop demand, and the Nash equilibria are *(Drop, Not drop), (Drop, Drop)*, or *(Not Drop, Drop)*. Further, the firm that does not drop initially will drop as a response. Therefore, the auction ends with no units sold.
2. If $P_a < 1$, both players have a strongly dominant strategy to choose *Not Drop*, leading the auction to close with payoff $1 - P_a > 0$ for both players (a duopoly).
3. If $1 \leq P_a \leq 2$, a mixed strategy equilibrium exists. Let $\pi_i^*((1,1),P_a)$ denote the probability of firm i choosing *Drop* at bid vector (1, 1), auction price P_a. Since the firms are symmetric, we have:

$$\pi_i^*((1,1),P_a)\ \frac{1}{2}\ (2-P_a) = \pi_i^*((1,1),P_a)\ (2-P_a) + (1-\pi_i^*((1,1),P_a))\ (1-P_a).$$

We solve this equation and find that $\pi_i^*((1,1),P_a) = 2 - \frac{2}{P_a}$. The expected payoff to either firm in this game is $V_i^*((1,1),P_a) = 3 - \frac{2}{P_a} + P_a$, which is maximised when $P_a = \sqrt{2}$, at which $V_i^*((1,1),\sqrt{2}) = 3 - \sqrt{8}$. Additionally, note that $V_i^*((1,1),\sqrt{8}-2) = V_i^*((1,1),\sqrt{2})$.

We graph the firms' expected payoffs as a dotted line in Fig. 1. The figure also shows the expected total surplus for the market, the auction's expected payoff, and an equally

[11] If both firms drop, we assign each firm 50% chance of dropping first, while allowing the second firm to respond, either by also dropping, or choosing not to drop.

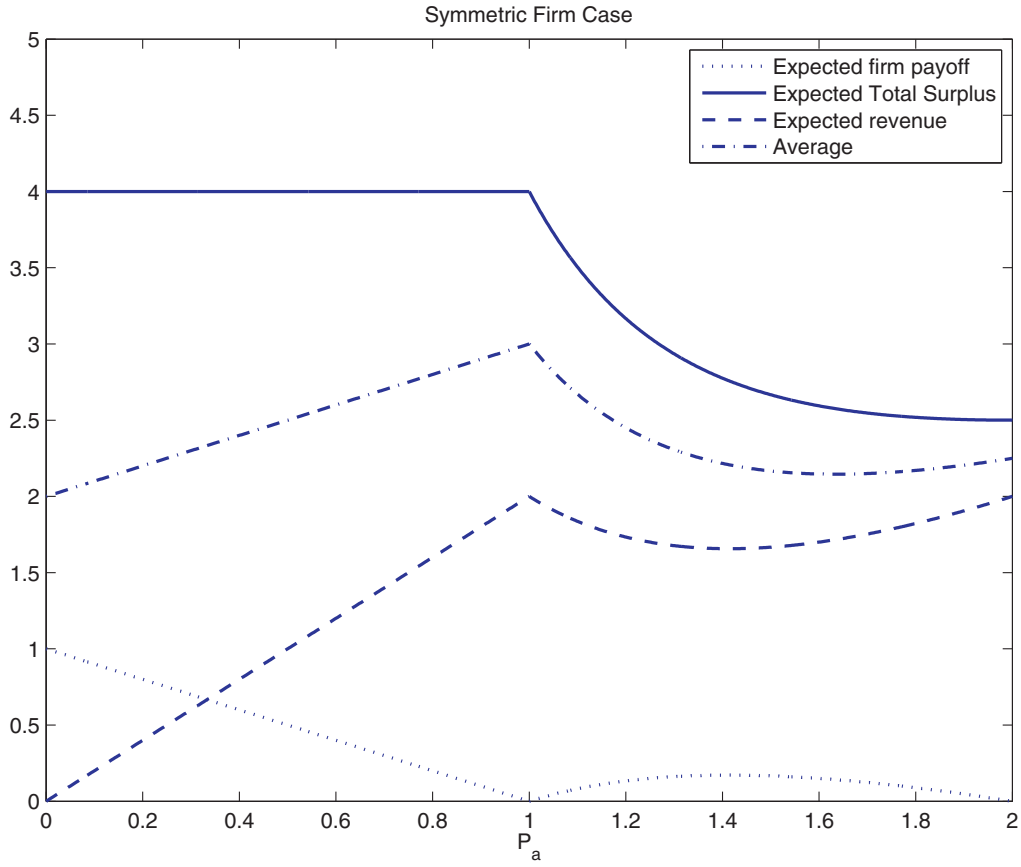


Fig. 1. Results from an auction of two units if the current bid vector is 1 unit each and firms have unit cost ($\mathbf{B} = (1, 1)$ if $c_1 = c_2 = 1$). The dotted line refers to the expected firm payoff as a function of the auction price. The solid and dashed lines represent the expected total surplus in the downstream market and the expected auction revenue, respectively. The dot-dashed line represents the average of these.

weighted average of these two to show how the game's outcome might be perceived by a regulator.

3.2. The (2,1) game

In this game, one firm has a current bid of 2 units, while the other bids only 1 unit. Table 3 illustrates firm payoffs. Unless $V_2^*((2, 1), P_a + \Delta P) < 0$, firm 2 has a weakly dominant strategy: *Not Drop*. Consequently, firm 1 can choose when to drop, which then ends the auction (once $\mathbf{B} = (1, 1)$, the auction will terminate, as described above).

How should firm 1 exercise this control? Given the payoff for each firm in the (1, 1) game, firm 1 will choose its dropping trigger to maximise this payoff function ($V_1^*((1, 1), P_a)$). Specifically, firm 1 drops if $P_a < \sqrt{8} - 2$ or $P_a > \sqrt{2}$. Firm 1 does not drop if $P_a \in (\sqrt{8} - 2, \sqrt{2})$. For the cases where indifference holds for the firms,

Table 3
The finite state game when both firms are low cost producers ($c_1 = c_2 = 1$), firm 1 is bidding 2 units, and firm 2 is bidding 1 unit ($\mathbf{B} = (2, 1)$). The auction price is P_a.

		Firm 2	
		Drop	Not drop
Firm 1	Drop	$\frac{1}{2}\max(2 - P_a, \frac{9}{4} - 2P_a, 0) + \frac{1}{2}V_1^*((1,1), P_a)$, $\frac{1}{2}V_2^*((1,1), P_a)$	$V_1^*((1,1), P_a)$, $V_2^*((1,1), P_a)$
	Not drop	$\max(2 - P_a, \frac{9}{4} - 2P_a, 0)$, 0	$V_1^*((2,1), P_a + \Delta P)$, $V_2^*((2,1), P_a + \Delta P)$

Table 4
Firm 1 expected payoff if it is bidding 2 units while firm 2 bids one unit ($\mathbf{B} = (2, 1)$). The table shows the firm's expected payoff ($V_1^*((2,1), P_a)$) as P_a varies. Table 3 shows the game that generates these results.

Firm 1 payoff	$P_a \leq \sqrt{8} - 2$	$\sqrt{8} - 2 < P_a \leq \sqrt{2}$	$\sqrt{2} < P_a \leq 2$	$2 < P_a$
$V_1^*((2,1), P_a)$	$1 - P_a$	$3 - \sqrt{8}$	$3 - \frac{2}{P_a} - P_a$	0

we assume that the probability of dropping or not dropping is 0.5 (see Section 4 and Appendix B.2).[12] Firm 1 obtains the payoffs displayed in Table 4 (also true for firm 2, given symmetry).

3.3. The (2,2) game

Given that both firms receive identical payoffs, as described in Table 4, neither firm will be concerned if its rival decides when to play the $\mathbf{B} = (1, 1)$ game. Hence we can conclude that there will be three Nash equilibria from $\mathbf{B} = (2, 2)$: *(Drop, Not Drop), (Drop, Drop)*, and *(Not Drop, Drop)*. If $\sqrt{2} > P_a > \sqrt{8} - 2$, then the firms will be indifferent between dropping and continuing, so there will be four Nash equilibria. Given this firm behaviour, we conclude that the auction will clear:

1. Immediately with a duopoly if the reserve price $\underline{P_a} < \sqrt{8} - 2$.
2. Either immediately (with a duopoly) or at $P_a = \sqrt{2}$ (with a duopoly or monopoly), if $\underline{P_a} = \sqrt{8} - 2$.
3. At $P_a = \sqrt{2}$ with a duopoly or monopoly, if $\sqrt{8} - 2 < \underline{P_a} < \sqrt{2}$.
4. Immediately with a monopoly or a duopoly if $\sqrt{2} \leq \underline{P_a} < 2$.
5. Immediately with a monopoly or no units sold if $\underline{P_a} = 2$.
6. With no units sold if $2 < \underline{P_a}$.

The top graph of Fig. 2 illustrates the outcomes of this auction, as a function of the reserve price $\underline{P_a}$. Using a cap of (2,2), (2,1), or (1,2) results in identical outcomes, since both firms agree about the optimal P_a at which to drop to $\mathbf{B} = (1, 1)$.

[12] Although the firms are indifferent, the regulator may not be indifferent between the two outcomes from the auction. For example, in this scenario, higher total surplus will be gained if the auction closes at $P_a = \sqrt{8} - 2$ (a duopoly) than if the auction closes at $P_a = \sqrt{2}$ (which may result in a monopoly). However, if the auction has a reserve price *just below* $\sqrt{8} - 2$, the auction will clear immediately, whereas for a reserve price *just above* $\sqrt{8} - 2$, the auction will clear at $P_a = \sqrt{2}$. The auction outcome (in terms of expected revenue or total welfare) if the reserve price is *exactly* $\sqrt{8} - 2$ will be the average of these two outcomes. This may result in an optimal auction design for the regulator involving a reserve price that is infinitesimally below or above a point of indifference for the firms.

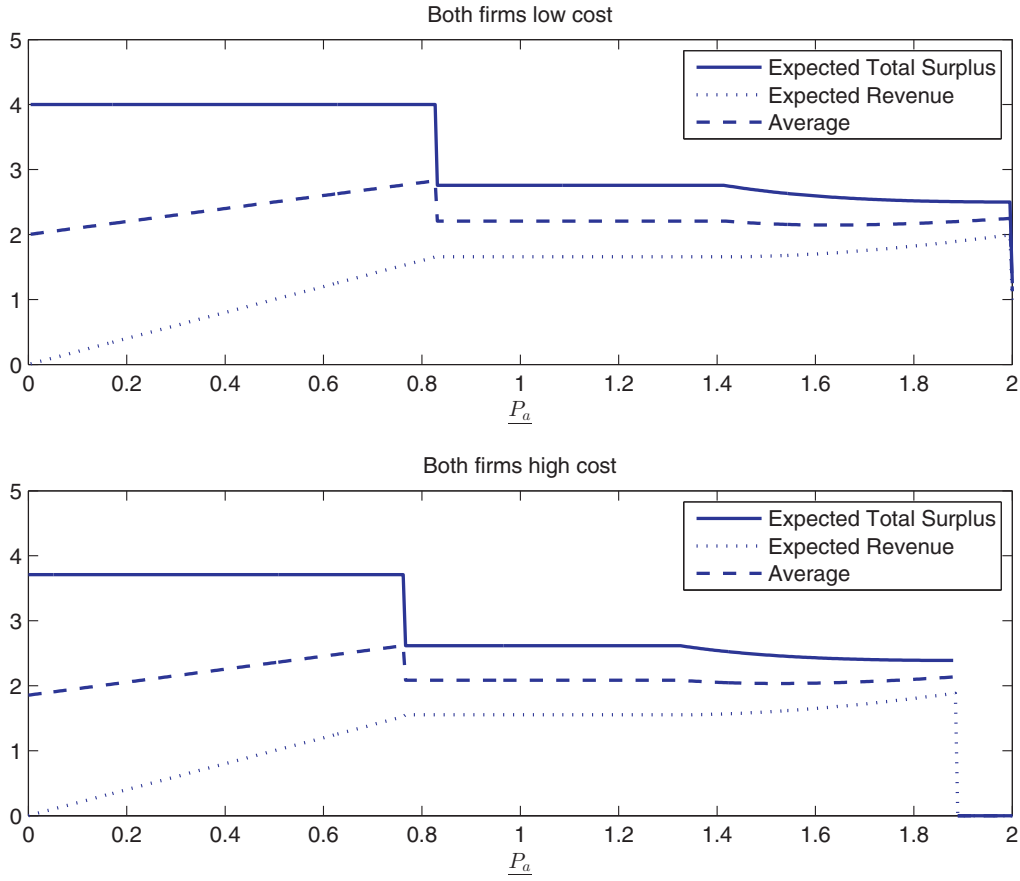


Fig. 2. Regulator expected payoffs for a two firm, two unit auction. These results are consistent for caps (2,2), (2,1), or (1,2). The top graph considers the case with both firms being low cost ($c_1 = c_2 = 1$). The bottom graph considers the firms being high cost ($c_1 = c_2 = \frac{10}{9}$), as discussed in Appendix A.1. In both cases, the horizontal axis measures reserve price, while the vertical axis measures the auction's outcome (in expected total surplus, expected revenue, or an average of the two).

We consider three measures of auction success, as noted earlier: expected total surplus, expected revenue, and an equally weighted combination of the two. By examining these graphs, a regulator could choose the optimal reserve price at which to start the auction.

Appendix A builds upon this section, by first analysing cases where firms have higher costs, and finally examining the behaviour of an uncertain regulator, who faces two firms, each of which has 50 percent probability of being high cost or low cost.

4. Solving the auction in general

We now turn our attention to the solution of an arbitrary auction (with many bidders and many units for sale). Given our solution to the downstream market and profit structure given in Eq. (4), we can calculate the payoffs that any firm receives at the end of

the auction. Suppose that the vector $\mathbf{B}$ is the firms' winnings at the end of the auction. Then, firm i's profit net of spectrum costs equals:

$$\begin{aligned}\overline{V}_i(\mathbf{B}, P_a) &= \Pi_i(q^*_{hi}, q^*_{li}, B_i, \mathbf{q^*_{h,-i}}, \mathbf{q^*_{l,-i}}, \mathbf{B_{-i}}, \mathbf{L}, P_a) \\ &= \Pi^D_i(q^*_{hi}, q^*_{li}, B_i, \mathbf{q^*_{h,-i}}, \mathbf{q^*_{l,-i}}, \mathbf{B_{-i}}, \mathbf{L}) - P_a\ B_i \end{aligned} \tag{7}$$

where the first term of Eq. (7) is the downstream profits, whereas the second term is the cost of the spectrum units to firm i. We will refer to the value function in Eq. (7) as the *terminal value* for firm i.

As illustrated for the two firm, two unit case in Section 3, we solve the auction as a dynamic programming problem (technical details are provided in Appendix B). To do this, we work through the bidding of the auction in reverse: determining how firms will behave at high prices to infer how they will behave at low prices.[13] We begin by considering a level of price sufficiently high that all firms would wish to exit the auction. We refer to this price as the *maximal price*. The market will thus clear at this stage, with all firms purchasing zero new spectrum units (and earning payoffs due to producing using their legacy spectrum holdings).[14]

As noted in Section 2.2, we assume that firms do not drop simultaneously. Thus if multiple firms choose to drop demand in the static game, each is equally likely to be the first to drop demand.

For each price/bid combination, we solve a finite state game (see Appendix B.2) to determine who lowers demand. This often takes the form of a game of congestion.[15] A firm would prefer not to lower demand if it knows its competitors will lower demand (and hence move the auction closer to its conclusion). However, if its competitors will not lower demand, it will be optimal for the firm to do so. These games are characterised by mixed strategy equilibria.

Equilibrium in the auction is described by the probability that each firm drops its demand, which we collectively denote by $\boldsymbol{\pi}^*(\mathbf{B}, P_a, \phi)$, given a particular combination of competitors' demands, its own demand, the current price, and which firm, if any, has just lowered demand (ϕ).

Our numerical solution technique relies upon working through the various bid level combinations in order, and from high prices to low prices, effectively solving the auction as a dynamic game. As noted earlier, we provide complete details of the solution technique in Appendix B.

[13] In contrast, if we were solving a Dutch (descending price) auction, we would solve the auction by considering what would happen if prices were *low* and use this behaviour to determine what would happen when prices are high (at the start of the auction).

[14] Maximal price is defined as the lowest price at which no firm would purchase any units, regardless of how many units its rivals won. For our examples in Sections 5 and 6, this means that the price exceeds the one-unit monopoly payoff.

[15] See Monderer and Shapley (1996) for discussion of congestion games.

Table 5
Parameters for the benchmark model (see Scenario 0 in Table 7). Demand for product j is given by $P_j = a_j + b_j \sum_{i=1}^{M} q_{ji}$, $j = h, l$. Costs are $C_{hi} = c_{hi} q_{hi}$ and $C_{li} = c_{li} q_{li}$.

Symbol	Value(s)	Explanation
N	9	Number of units auctioned
M	3	Number of firms
a_h	13	Choke price in the high-product market
a_l	9	Choke price in the low-product market
b_h	1	Slope of the demand curve for the high product
b_l	1	Slope of the demand curve for the low product
$\mathbf{c}_h$	(1, 1, 1)	Marginal cost of producing the high product
$\mathbf{c}_l$	(1, 1, 1)	Marginal cost of producing the low product
$\mathbf{B}_l$	(2, 2, 2)	Endowment of legacy spectra
ΔP	0.1	Price increment in auction

5. Results for uncapped auctions with no reserve price

As a numerical example, we focus on a scenario whose parameters are given in Table 5. Some comments regarding this choice of numbers are in order. We have chosen to consider an auction with $M = 3$ firms over $N = 9$ units. Three firms are sufficient to avoid the singular feature of the two firm case: that spectra not won by a firm are necessarily either won by its one competitor or unused. With three firms, a firm who does not buy a particular spectrum unit may be uncertain as to which of its competitors will win the unit. The nine-unit auction is also interesting, because it allows a symmetric allocation of units. Mixed strategies are thus a function of firm behaviour and not an artefact of auction asymmetry.

With three firms and nine units, there are 1000 bid combinations at a given price. We assume \$0.1 price increments from \$0 to \$11; this generates 110,000 combinations and 440,000 finite state games when we condition on which firm dropped its bid last.[16]

With our set of base case parameters, we consider nine scenarios that provide some insight into the workings of a spectrum auction. Each scenario is a slight variation of the benchmark model (Scenario 0), which assumes parameter values given in Table 5. These scenarios are summarised in Table 6. Scenarios 0–3 focus on the effect of cost variation in a triopoly market. A common asymmetry in spectrum auctions is the interaction between incumbents and entrants (Cramton et al., 2011). Therefore, Scenario 4 and 5 focuses on a duopoly that faces entry by a single firm; this entrant is a high (low) cost producer in the high product market. Finally, Scenarios 6–8 investigate the effect of two potential high/low cost entrants into a monopoly market.

Because of mixed strategies for bid lowering, a given auction's outcome will be (in general) a random variable. Hence our solutions take the form of probability distributions

[16] We choose an upper bound of \$11, since this is sufficiently high that even if its competitors were bidding 0 units, a firm would not buy a single spectrum unit.

Table 6
Scenarios considered in Section 5. These are modifications of the benchmark model described in Table 5. The outcomes from these auctions are displayed in Table 7.

Scenario	Characteristics
0	Benchmark model: all firms are incumbents and low cost.
1	Firm 1 is high cost: $c_{h1} = 3$.
2	Firms 1 and 2 are high cost: $c_{h1} = c_{h2} = 3$.
3	All three firms are high cost: $c_{h1} = c_{h2} = c_{h3} = 3$.
4	Firm 1 is an entrant: $L_1 = 0$.
5	Firm 1 is a high cost entrant: $L_1 = 0, c_{h1} = 3$.
6	Firms 1 and 2 are entrants: $L_1 = L_2 = 0$.
7	Firms 1 and 2 are entrants and firm 1 is high cost: $L_1 = L_2 = 0, c_{h1} = 3$.
8	Firms 1 and 2 are high cost entrants: $L_1 = L_2 = 0, c_{h1} = c_{h2} = 3$.

over prices per unit of spectrum, spectra allocated to firms, and firm profits in the downstream market.

The results are found in Table 7, including the price and allocation of the spectra at the end of the auction, the price and output produced in the high product market, as well as the total surplus and auction revenue.

Scenario 0 presents our base case: all firms are incumbents and symmetric. The average spectra obtained by any firm is 2.59 units, while the auction clears at an average price of \$1.04 per unit. If sold, the spectrum units are almost always fully utilised (firms are producing the maximum number of high product units that their winnings permit), so the total output in the high product market is generally 8 units, although there is some probability of the auction clearing at \$4, with less units being sold. Notice that since the firms are symmetric, the average spectra won, and hence profit, is identical across the three firms. Although the solution is symmetric in expectation, many auction outcomes result in asymmetric allocations of spectra, with one firm obtaining at least 4 spectrum units and holding market power.

Eső et al. (2010) show that (using a VCG auction) a capacity auction with greater surplus capacity can lead to "hoarding" by firms. This results in some firms having surplus capacity, while others are more constrained in their downstream production than would be the case in a more austere auction.

In our model, since we consider a uniform price clock auction, firms do not necessarily have to buy all the units at auction. Indeed, in many cases, when the price is high, firms may not want to buy additional units. We rarely see hoarding in our model. However, similar to Eső et al. (2010), we do observe the auction resulting in constraints on firm output, even when there are sufficient units of spectrum available for all firms to produce their unconstrained Cournot outputs (see Table 8). For example, in Scenario 0, the three firms would each produce 3 units of high product outputs, if their capacity was not constrained. With 9 spectrum units available, this is possible, with each firm winning 3 spectrum units at zero auction price. However, since the marginal benefit of the extra unit is high when the auction price is zero, the firms actively compete for more units.

Table 7
Results for scenarios in Table 6. Rows show outcomes with probabilities (probabilities below 0.02 suppressed). For scenarios 0, 1, 3, 4, and 5, final row shows average. Columns are winnings by firm j (B_j), auction closing price (P_a), firm j high product quantity (q_{hj}), high product price (P_h), total surplus (TS), and revenue (Rev).

Scenario	Probability	B_1	B_2	B_3	P_a	q_{h1}	q_{h2}	q_{h3}	P_h	TS	Rev
0	0.12	2	2	4	0.8	2	2	4	5	94	6.4
0	0.12	4	2	2	0.8	4	2	2	5	94	6.4
0	0.12	2	4	2	0.8	2	4	2	5	94	6.4
0	0.07	4	2	2	0	4	2	2	5	94	0
0	0.07	5	2	2	0	4	2	2	5	94	0
0	0.07	2	4	2	0	2	4	2	5	94	0
0	0.07	2	5	2	0	2	4	2	5	94	0
0	0.07	2	2	4	0	2	2	4	5	94	0
0	0.07	2	2	5	0	2	2	4	5	94	0
0	0.02	2	2	1	4	2	2	1	8	77.5	20
0	0.02	2	1	2	4	2	1	2	8	77.5	20
0	0.02	1	2	2	4	1	2	2	8	77.5	20
0	Avg.	2.59	2.59	2.59	1.04	2.52	2.52	2.52	5.44	91.64	6.62
1	0.25	1	4	3	0	1	4	3	5	92	0
1	0.25	1	5	3	0	1	4	3	5	92	0
1	0.25	1	3	4	0	1	3	4	5	92	0
1	0.25	1	3	5	0	1	3	4	5	92	0
1	Avg.	1	3.75	3.75	0	1	3.5	3.5	5	92	0
2	1	2	2	2	1.1	2	2	2	7	76	6.6
3	0.15	2	2	1	2	2	2	1	8	67.5	10
3	0.15	2	1	2	2	2	1	2	8	67.5	10
3	0.15	1	2	2	2	1	2	2	8	67.5	10
3	0.07	3	2	1	2	3	2	1	7	72	12
3	0.07	3	1	2	2	3	1	2	7	72	12
3	0.07	2	3	1	2	2	3	1	7	72	12
3	0.07	1	3	2	2	1	3	2	7	72	12
3	0.07	2	1	3	2	2	1	3	7	72	12
3	0.07	1	2	3	2	1	2	3	7	72	12
3	0.02	1	1	2	4.1	1	1	2	9	62	16.4
3	0.02	2	1	1	4.1	2	1	1	9	62	16.4
3	0.02	1	2	1	4.1	1	2	1	9	62	16.4
3	Avg.	1.79	1.79	1.79	2.23	1.79	1.79	1.79	7.64	68.98	11.71
4	0.33	1	4	1	1.8	1	4	1	7	78	10.8
4	0.33	1	1	4	1.8	1	1	4	7	78	10.8
4	0.17	1	2	1	5.2	1	2	1	9	64	20.8
4	0.17	1	1	2	5.2	1	1	2	9	64	20.8
4	Avg.	1	2.17	2.17	2.93	1	2.17	2.17	7.67	73.33	14.13
5	0.5	1	4	3	1	0.33	3.83	3	5.83	86.64	8
5	0.5	1	3	4	1	0.33	3	3.83	5.83	86.64	8
5	Avg.	1	3.5	3.5	1	0.33	3.42	3.42	5.83	86.64	8
6	1	1	1	7	0	1	1	5	6	83.5	0
7	1	1	2	6	0	0.25	1.75	4.75	6.25	82.69	0
8	1	1	1	7	0	0.5	0.5	5.5	6.5	80.75	0

Table 8
Downstream market production decisions for 3 firms assuming *no constraints*. This shows the Cournot equilibrium for monopoly, duopoly, and triopoly production in the benchmark model (see Table 5). We examine each market separately, since, without constraints, the markets can be considered in isolation. Note that for the triopoly case, with low costs, exactly 9 high products are output, along with 2 low products. This would (in a constrained setting) require use of 9 units of new spectrum. When the costs are high, 7.5 high products are output, along with 2 low products. This would (with constraints) require 7.5 units of new spectrum (less than what is available in the auction).

Outcome	Low cost firms				High cost firms			
	$c_1 = c_2 = c_3 = 1$				$c_1 = c_2 = c_3 = 3$			
	q_{hi}	TS_h	q_{li}	TS_l	q_{hi}	TS_h	q_{li}	TS_l
Monopoly[a]	6	54.000	4.00	24.000	5.00	37.500	4.00	24.000
Duopoly[b]	4	64.000	2.67	28.444	3.33	44.444	2.67	28.444
Triopoly[c]	3	67.500	2.00	30.000	2.50	46.875	2.00	30.000

[a] The *monopoly* outcome refers to a case where only one firm produces in the given market.
[b] The *duopoly* outcome refers to a case where two firms produce in the given market.
[c] The *triopoly* outcome refers to a case where all three firms produce in the given market.

According to the results of Scenario 0, there is only a low probability (0.5%) of each firm winning 3 spectrum units.

In Scenario 1, firm 1 has higher cost than firms 2 and 3. The auction results in four possible outcomes, in all of which one of the two low cost firms (i.e., firms 2 and 3) obtains 4–5 units and holds market power, while firm 1 wins 1 unit and the other low cost firm wins 3 units. Consequently, firms 2 and 3 produce more, on average, than firm 1. In all cases, the auction closes at price zero, resulting in zero revenue.[17]

In Scenario 2, where firms 1 and 2 are high cost producers in the high product market, we see that high cost producers can hold up the market, preventing the low cost firm from obtaining more units. The main reason for this outcome is the trade-off for the low cost firm. On the one hand, it can increase the price to obtain more units and force the high cost firms to either exit the market or settle for less units. On the other hand, doing so increases the spectrum price, which offsets its profits. Conversely, the two high cost firms can credibly threaten to increase the price, since they will each be less hurt by this than the low cost firm, because they will win less units. Consequently, all firms win 2 units at the auction price of \$1.1 which results in high revenue (\$6.6).

Scenario 3 has 3 symmetric but high cost producers, obtaining 1.79 units on average. Total surplus is the lowest of all the cases so far (reflecting the high cost industry), but the auction is very lucrative, with average revenue of \$11.71. Our scenario 3 also echoes the behaviour described in Eső et al. (2010). Even though 9 units are more than sufficient to supply the market (see Table 7), the firms bid aggressively, and as a result, in many cases, firms are constrained in their downstream production.

[17] At a price of zero, it might seem that firms would necessarily have infinite demand for units. However, due to the dynamic nature of the game, if firms choose bid levels such that there is excess demand, the price will rise to being non-zero, which would result in lower payoffs for firms than if they bid in such a way as to create no excess demand.

Scenarios 4–5 cover the case where one firm (firm 1) is an entrant, so it has no legacy spectra. Since firm 1 could use spectra both in the low and high product markets, this provides an incentive for firms 2 and 3 to bid aggressively in the auction. In both cases, the incumbent firms 2 and 3 increase the auction price to discourage firm 1 from obtaining more than one unit, which firm 1 then utilises solely in the high product market. In Scenario 5 (where the entrant has high cost), the price is lower, allowing the low cost incumbents to obtain more units and increasing total surplus. Thus, the entrant ends up utilising its 1 unit for both high and low products.

Scenarios 6–8 focus on the cases where there are two potential entrants (firms 1 and 2). In all these scenarios, the auction clears at zero price. If both entrants are low cost producers (Scenario 6), then firm 3 can effectively prevent either from obtaining a second unit, thus dominating the market with 7 units. If only one entrant is low cost (firm 2 in Scenario 7), the low cost entrant can obtain 2 units, while the incumbent produces 4.75 high product units with its 6 spectrum units (purchased to shut out firms 1 and 2). When both entrants are high cost (Scenario 8), the incumbent is able to ensure each entrant wins only 1 unit. Contrasting Scenario 7 to Scenario 6, entrant firm 2 has stronger incentives to bid for a second unit when one competitor (firm 1) is high cost.

In summary, results of our model are consistent with Eső et al. (2010), in that, even though firms could share the spectra equally at zero auction price, they choose to actively compete in the auction to get extra units. This results in many outcomes where one firm wins more than the others, even if the firms are ex-ante symmetric (Scenarios 0 and 3).

Conventional wisdom (from single item English auctions) implies that increased competition in an auction is unambiguously good for revenue and total surplus maximisation (because only the second highest bid is relevant). In the multi-unit case, bidder composition matters. In our Scenario 2, high cost producers hold up the market. Given that two low cost firms can effectively limit a single high cost firm's production (see Scenario 1), we conclude that the ratio of high to low cost firms is important in the auction. From an auctioneer's perspective, this suggests that it will (under some circumstances) be beneficial, for social welfare outcomes, to preclude high cost bidders from participating in an auction.

6. Regulation through caps and reserve prices

Although an auctioneer generally aims to maximise revenue in an auction, a regulator tasked with running a spectrum auction may also take into account total surplus in the downstream market generated through the allocation of spectra. In particular, the regulator may face a trade-off between the auction revenue and total surplus; see Cramton et al. (2011). For example, in Scenario 2 of Section 5, a regulator may be concerned that the high cost firms bid so as to prevent the low cost firm from winning more units. The regulator may decide to intervene in the auction by setting a reserve price and caps to prevent this behaviour.

We note that while we use the term "regulator" here, the auctioneer could, in many alternative applications, be a profit maximising firm. For example, in auctioning airport gates, the airport may impose a capping policy, and may be concerned about both auction revenue and airport passenger throughput (quantity produced), since this will affect retail rental revenues.

We assume that the regulator is uncertain as to which firms can exploit spectra at low cost, even though firms know each other's type in the auction. For example, each firm may have a different technology available to use in the high product market. In focusing on uncertainty, we omit cases where firm type is known to the regulator with certainty. In these cases, a regulator can generally maximise total surplus by assigning units to firms, and the information revelation side of the auction is unnecessary.

If the regulator is uncertain regarding demand/supply parameters, then, conditional on $\overline{\mathbf{B}}$ and $\underline{P}_a$, we can evaluate the moments of total surplus (TS) and revenue (R) as follows:

$$\mathbb{E}(X) = \sum_{\omega\in\Omega} \psi_\omega \mathbb{E}(X_\omega) \tag{8a}$$

$$\mathbb{V}(X) = \sum_{\omega\in\Omega} \psi_\omega \left[\mathbb{V}(X_\omega) + \mathbb{E}(X_\omega)^2\right] - \mathbb{E}(X)^2 \tag{8b}$$

$$\mathrm{Cov}(TS, R) = \sum_{\omega\in\Omega} \psi_\omega [\mathrm{Cov}(TS_\omega, R_\omega) + \mathbb{E}(TS_\omega)\mathbb{E}(R_\omega)] - \mathbb{E}(TS)\mathbb{E}(R), \tag{8c}$$

where $\mathbb{E}(\cdot)$ and $\mathbb{V}(\cdot)$ represent the conditional mean and the conditional variance of the relevant variables, and $X = TS,\ R$. The set of possible states that demand and supply parameters may take is denoted by Ω, and ψ_ω is the probability that state $\omega \in \Omega$ occurs.[18]

The regulator optimally chooses a cap structure and a reserve price for the auction depending on his/her objective function, defined as follows:

$$\max_{\overline{\mathbf{B}}\geq 0, \underline{P}_a \geq 0} \mathbb{E}\big(\alpha_1\ TS + \alpha_2\ R \mid \overline{\mathbf{B}}, \underline{P}_a\big) + \alpha_3\ \mathbb{V}\big[\alpha_1\ TS + \alpha_2\ R \mid \overline{\mathbf{B}}, \underline{P}_a\big]. \tag{9}$$

The first term in (9) is a weighted average of the expected total surplus and revenue, and the last term is the variance of this weighted sum. The expectation and variance are evaluated using (8). This objective function generalises the one used in Dana and Spier (1994), and allows the regulator to be risk-neutral (when $\alpha_3 = 0$) or risk-averse (when $\alpha_3 < 0$).[19] We solve this maximisation problem by working through each cap/reserve combination, since there are a small range of caps that can be set (e.g., for a 9 unit auction, there are 1000 possible caps for three firms) along with a finite number of reserve prices (e.g., 110 price levels based on maximal price of 11 and price increment

[18] Note that TS_ω and R_ω are both (in general) random variables, since, as per the examples in Section 5, even with full information, there will mixing in the equilibrium strategies which results in multiple outcomes.
[19] The regulator could be risk-averse either due to societal preferences, or because the regulator has his/her performance/compensation measured based upon auction outcomes.

of 0.1).[20] Constraints, such as non-discriminatory capping, can be included by reducing the set of caps considered.[21]

In our examples that follow, we consider six different regulatory objective functions. The first three regulators are all risk-neutral ($\alpha_3 = 0$), and have different objectives regarding total surplus and revenue. Regulator 1 is concerned only with total surplus maximisation. Regulator 2 is concerned only with revenue maximisation, while regulator 3 puts equal weights on total surplus and revenue. Our second three regulators are all risk-averse and penalised according to the variance of their outcomes ($\alpha_3 = -1$), but have similar objectives to their risk-neutral counterparts. Regulator 4 maximises total surplus, regulator 5 maximises revenue, and regulator 6 maximises the sum of total surplus and revenue.

Using our results from Section 5, we explore three scenarios. In the first scenario (see Table 9), three incumbent firms compete in the market. Each firm has a 50% probability of being low cost ($c_{hj} = 1$) or high cost ($c_{hj} = 3$). Our second example features two incumbents (firms 2 and 3), who are known to be low cost, facing potential entry by a third firm (firm 1). The entrant has a 50% probability of being either high or low cost. Here, the regulator faces a tradeoff: on the one hand, encouraging firm 1 to enter the market could improve market efficiency. On the other hand, if the entrant is a high cost firm, this may result in higher prices in the market (reducing total surplus). The final example features a single incumbent (firm 3; known to be low cost) and two potential entrants (firms 1 and 2; each with 50% probability of being low cost).[22]

Examining Table 9, the most striking feature is the paucity of auctions with more than one round of bidding. For all of the three incumbent cases, the optimal caps sum to less than or equal to nine (the total supply), indicating that the auction will clear at the reserve price. Uncertainty regarding revenue still prevails, since firms may choose to buy less than the quantity offered to them by the regulator. It is interesting to note that even for revenue maximising regulators (regulators 2 and 5), offering less than the full quantity to auction is optimal. This is because bidders may value units higher if they know that their competitors cannot purchase more units. Hence more revenue can be earned by selling 6 units than selling 9.

It is also worth noting that a tradeoff between total surplus and revenue exists. For regulators 1 and 4, who maximise total surplus, revenue is equal to 0. In contrast, the

[20] When reporting our results, we show the least restrictive optimal cap/reserve price structure, if the regulator is indifferent between several combinations. To define this, we consider a cap/reserve combination to be less restrictive if it has a higher or equal cap for all participants, with the same or lower reserve price (with one equality holding strictly). Once more restrictive caps/reserves are eliminated, we often obtain a unique cap/reserve combination. Where we do not, we report all the combinations that are not eliminated. This results in some cases where two or three symmetric alternatives are presented (see, for example, regulator 3 in the three incumbent case). Graphs of the regulators' utility from alternative cap/reserve choices are available on request from the authors.

[21] Appendix A.3 contains an example of an uncertain regulator choosing a cap and reserve price policy for the two firm, two unit case of Section 3.

[22] Parameters are as in Section 5, so incumbents have $L_i = 2$, entrants have $L_i = 0$, all firms have $c_{li} = 1$, etc.

Table 9
Optimal cap and reserve price for regulated market. $\overline{\mathbf{B}}$ ($\underline{P}_a$) reports the cap (reserve) for the least restrictive cap/reserve combination that will maximise utility for the regulator. B_j shows the mean winnings by firm j, with standard deviation reported in parentheses. Similarly, TS (Rev) reports the mean total surplus (revenue) earned by this regulator, with standard deviation in parentheses. Lastly, $Cov(TS, Rev)$ reports the covariance of total surplus and revenue. Regulators 1–3 are risk-neutral, with regulator 1 maximising total surplus ($(\alpha_1, \alpha_2, \alpha_3) = (1, 0, 0)$), regulator 2 maximising revenue ($(\alpha_1, \alpha_2, \alpha_3) = (0, 1, 0)$), and regulator 3 maximising an average of the two ($(\alpha_1, \alpha_2, \alpha_3) = (0.5, 0.5, 0)$). Regulators 4–6 are risk-averse, with regulator 4 maximising total surplus ($(\alpha_1, \alpha_2, \alpha_3) = (1, 0, -1)$), regulator 5 maximising revenue ($(\alpha_1, \alpha_2, \alpha_3) = (0, 1, -1)$), and regulator 6 maximising an average of the two ($(\alpha_1, \alpha_2, \alpha_3) = (0.5, 0.5, -1)$).

3 incumbents case where the probability of low cost is (0.5, 0.5, 0.5)								
Regulator	1	2	3	3	3	4	5	6
$\overline{\mathbf{B}}$	(3,3,3)	(2,2,2)	(3,3,2)	(3,2,3)	(2,3,3)	(2,2,2)	(1,1,1)	(2,2,2)
$\underline{P}_a$	0	4.8	2.9	2.9	2.9	0	6.9	2.9
B_1	2.94 (0.09)	1.58 (0.37)	2.23 (0.67)	2.23 (0.67)	1.65 (0.38)	2 (0)	1 (0)	2 (0)
B_2	2.94 (0.09)	1.58 (0.37)	2.23 (0.67)	1.65 (0.38)	2.23 (0.67)	2 (0)	1 (0)	2 (0)
B_3	2.94 (0.09)	1.58 (0.37)	1.65 (0.38)	2.23 (0.67)	2.23 (0.67)	2 (0)	1 (0)	2 (0)
TS	86.78 (5.93)	71.33 (7.14)	79.22 (6.98)	79.22 (6.98)	79.22 (6.98)	78 (3.46)	58.5 (1.73)	78 (3.46)
Rev	0 (0)	22.76 (3.33)	17.69 (2.41)	17.69 (2.41)	17.69 (2.41)	0 (0)	20.7 (0)	17.4 (0)
$Cov(TS, Rev)$	0	22.58	13.88	13.88	13.88	0	0	0
2 incumbents case where the probability of low cost is (0.5, 1, 1)								
Regulator	1	1	2	3	4	4	5	6
$\overline{\mathbf{B}}$	(0,9,4)	(0,4,9)	(2,2,2)	(3,3,3)	(0,9,4)	(0,4,9)	(0,2,2)	(2,3,3)
$\underline{P}_a$	0	0	6.9	2.4	0	0	6.9	2.4
B_1	0 (0)	0 (0)	0.7 (0.73)	2.59 (0.36)	0 (0)	0 (0)	0 (0)	2 (0)
B_2	5 (0)	4 (0)	1.7 (0.34)	2.77 (0.42)	5 (0)	4 (0)	2 (0)	3 (0)
B_3	4 (0)	5 (0)	1.7 (0.34)	2.77 (0.42)	4 (0)	5 (0)	2 (0)	3 (0)
TS	91.31 (0)	91.31 (0)	64.65 (2.28)	86.31 (4.34)	91.31 (0)	91.31 (0)	64 (0)	86.38 (1.37)
Rev	0 (0)	0 (0)	28.2 (2.15)	19.5 (1.85)	0 (0)	0 (0)	27.6 (0)	19.2 (0)
$Cov(TS, Rev)$	0	0	4.91	7.61	0	0	0	0
1 incumbent case where the probability of low cost is (0.5, 0.5, 1)								
Regulator	1	1	2	3	3	4	5	6
$\overline{\mathbf{B}}$	(9,2,9)	(2,9,9)	(3,3,2)	(4,2,3)	(2,4,3)	(4,4,7)	(2,2,3)	(3,3,3)
$\underline{P}_a$	0	0	4.4	2.9	2.9	0	3.9	2.4
B_1	1.75 (1.69)	0.75 (0.19)	2.41 (0.57)	3.63 (0.35)	1.92 (0.13)	1.25 (0.19)	2 (0)	3 (0)
B_2	0.75 (0.19)	1.75 (1.69)	2.41 (0.57)	1.92 (0.13)	3.63 (0.35)	1.25 (0.19)	2 (0)	3 (0)
B_3	6.5 (0.76)	6.5 (0.76)	1.5 (0.50)	2.86 (0.24)	2.86 (0.24)	6.5 (0.25)	3 (0)	3 (0)
TS	82.49 (1.09)	82.49 (1.09)	65.75 (8.48)	77.43 (5.86)	77.43 (5.86)	82.41 (1.01)	71 (2.08)	80 (2.78)
Rev	0.16 (0.71)	0.16 (0.71)	27.86 (5.02)	24.39 (2.38)	24.39 (2.38)	0 (0)	27.3 (0)	21.6 (0)
$Cov(TS, Rev)$	0.12	0.12	40.98	12.86	12.86	0	0	0

actions of regulators maximising revenue or a combination objective (regulators 2, 3, 5, and 6) result in lower total surplus.

Although the revenue maximising regulators run auctions clearing at the reserve price immediately, they extract revenue by offering a high reservation price ($4.8 for regulator 2 and $6.9 for regulator 5). The risk entailed in this strategy is that firms may be high cost, and decline to purchase all of the units they are offered. Hence once risk-aversion is included (regulator 5), the revenue maximising regulator offers only 3 units for sale, ensuring that all are sold. In contrast, total surplus maximising regulators (regulators 1 and 4) tend to offer higher quantities for auction.

The mixed objective regulators (regulators 3 and 6) have cap/reserve combinations that are a convex combination of their total surplus and revenue maximising counterparts. For risk-neutral mixed objectives (regulator 3), two of the firms are offered a cap of 3, while the third firm is offered 2. This more austere cap combination allows a reserve price of $2.9, which leads to an average revenue of $17.69 while reducing expected total surplus by only $7.56 relative to the (3,3,3) cap offered by regulator 1. Similarly, regulator 6 (risk-averse mixed objective) offers 2 units to each firm, with a reserve price of $2.9. This earns $3.3 less revenue than that earned by regulator 5 (risk-averse revenue maximiser) but achieves the same social welfare as regulator 4 (risk-averse total surplus maximiser).

When two of the firms are incumbents, and the third firm an entrant (with uncertain costs), the auction ends immediately at the reserve price. Regulators 1 and 4 cap one incumbent at 9 units and the other at 4. In this case, the auction clears immediately, with one incumbent buying 5 units and the other 4. Since there is no uncertainty surrounding the incumbents, this strategy is risk-free. In the case of total surplus maximisation, having two low cost firms forming a duopoly is preferable to a triopoly. This is due to the risk (in a triopoly) of the entrant proving to be high cost. Regulator 2 offers 2 units to each firm, gambling that the entrant will be low cost, which increases competition and so the revenue. In contrast, regulator 5 only offers 2 units to each of the *incumbents*. This eliminates risk, but at the price of $0.6 lower expected revenue. The mixed objective risk-neutral regulator offers each firm 3 units (at reserve $2.4), while its risk-averse counterpart offers 2 units to the entrant and 3 to each incumbent. Again, favouring the (known) incumbents considerably reduces risk. Regulator 6 is the only risk-averse regulator who is willing to offer units to the entrant. This behaviour is consistent with our conclusion from Section 5: in the face of (potentially) high cost bidders, it may be optimal to limit their participation in the auction.

For the case of two entrants, we see some actual auctions proving optimal. This case represents a difficult one for total surplus maximisers. Favouring the incumbent runs the risk of a monopoly, with low total surplus, but on the other hand, there is considerable uncertainty regarding the entrants' costs. Here regulator 1 runs an auction where the incumbent and one entrant have caps of 9 and the other entrant has a cap of 2. Aggressively capping one entrant makes it more likely to see the other (uncapped) entrant win a second unit, resulting in a more competitive downstream market.

Although regulator 4 offers caps of (4,4,7), the auction clears immediately (at price 0) with the distribution depending on the costs of the firms. When revenue is of primary concern, the treatment of caps depends upon risk aversion. Regulator 2 offers caps of 3 to the entrants and 2 to the incumbent, whereas regulator 5 chooses a more cautious strategy of offering caps of 2 to entrants and 3 to the incumbent. This more cautious strategy earns less revenue on average, but achieves higher total surplus. Lastly, the mixed objective cases choose quite balanced caps, with regulator 3 offering caps of 4 and 2 to the entrants, and 3 to the incumbent. Regulator 6 offers a cap of 3 to all firms. In both cases, modest reserve prices are charged.

We draw several conclusions from these scenarios. First, a multiple round auction is a useful tool only for total surplus maximisation when the concerns with uncertainty in production efficiency are high relative to market power concerns (see the two entrants case). In most other cases, a single round auction with aggressive caps and reserve price achieves better outcomes than running a multiple round auction with a lower reserve or higher caps.

Second, when a regulator is risk-averse, it results in lower caps and higher reserve price in the three and two incumbents cases, and in most cases a two- or three-fold reduction in the variation of the total surplus and revenue.

Third, caps are especially useful to increase total surplus (with or without uncertainty). In contrast, when revenue maximisation is the objective, reserve prices are a more effective tool for extracting surplus. It is interesting to note that in this case, the regulator restricts the supply of units to increase its revenue: effectively acting as a monopolist in the spectrum unit market. We remark that, empirically, auctions closing at their reserve price may be an indication of a revenue maximising auctioneer.

7. Conclusion

In this paper, we explore a multi-unit auction in which market participants use the auction units to provide services in a downstream market. Winning large quantities in the auction allows a participant to wield considerable power in the downstream market. Although our model is couched in the application to the Digital Dividend auction, it can be used to examine other auction situations (with discrete actions and full information by bidders) such as bidding for gates at an airport, or the auctioning of physical commodities.

Our findings for un-capped auctions suggest a relationship between the mix of firm costs and the revenue/total surplus generated by the auction. When low cost firms dominate the market (two low cost firms, and one high cost firm), the two low cost firms allow the high cost firm to enter the market. In doing so, the auction clears at zero price, so the spectra are free of charge for all firms. Consequently, total surplus is high, while revenue stays low. On the other hand, when the mix is reversed (two high cost firms, and one low cost firm), the two high cost firms can hold up the market, winning as many units as the single low cost firm. In doing so, they threaten to increase the auction price, making it very costly for all firms. This raises the auction revenue, while lowering total

surplus. This finding suggests that the mix of participants matters, and allowing more (high cost) bidders may lead to worse auction outcomes.

Similar to Eső et al. (2010), we show that in an auction with plenty of spectrum units available, aggressive bidding by firms can lead to a constrained downstream equilibrium.

Lastly, we show how the model can be generalised to consider a risk-neutral/averse regulator managing the auction, who is maximising social welfare in the downstream market and/or auction revenue. Risk aversion also has important effects on auction outcomes, with risk averse auctioneers achieving lower expected profit and total surplus. Using the capping structures and the reserve price, we find that an auction is particularly useful when the regulator maximises total surplus. Meanwhile, a regulator maximising auction revenue sets reserve prices high enough to restrict the units sold, to increase its revenue.

One possible extension to this paper would be to consider multiple periods in the downstream market. Here the complication added would be that future growth in demand for the products could lead to spectrum units that are initially unused becoming utilised in later instances of the market clearing. A second possible extension would be to extend the set of policy tools available to the regulator. For example, subsidies to firms in the form of a reduced price per spectrum unit won could be considered. In this case, select firms might be encouraged to bid more aggressively, at the cost of lower auction revenues due to the subsidy.

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Appendix A. The 2-unit, 2-firm example: full analysis

The example contained in the main body of the text considers two symmetric firms bidding for two units of spectrum. In this Appendix, we consider the case of firms having higher costs, having asymmetric costs, and finally a regulator making a decision as to how to use caps and reserve prices to run an auction. Payoffs in the downstream market for these cases (along with the low cost case discussed in Section 3) are summarised in Table 1.

Table 10
The (1,1) game when both firms are high cost producers.

		Firm 2	
		Drop	Not drop
Firm 1	Drop	$\frac{1}{2}\max\left(\frac{17}{9}-P_a,0\right)$, $\frac{1}{2}\max\left(\frac{17}{9}-P_a,0\right)$	0, $\max\left(\frac{17}{9}-P_a,0\right)$
	Not drop	$\max\left(\frac{17}{9}-P_a,0\right)$, 0	$\left(\frac{26}{27}\right)^2-P_a$, $\left(\frac{26}{27}\right)^2-P_a$

Table 11
The (1,1) game when firm 1 is a low cost producer and firm 2 is high cost.

		Firm 2	
		Drop	Not drop
Firm 1	Drop	$\frac{1}{2}\max\left(2-P_a,0\right)$, $\frac{1}{2}\max\left(\frac{17}{9}-P_a,0\right)$	0, $\max\left(\frac{17}{9}-P_a,0\right)$
	Not drop	$\max\left(2-P_a,0\right)$, 0	$\frac{19}{18}-P_a$, $\left(\frac{17}{18}\right)^2-P_a$

A1. Case 2: both firms are high cost

We repeat the same procedure from Section 3 to examine the case where both firms have high marginal costs: $c_1 = c_2 = 10/9$. The second block (columns 4 and 5) in Table 1 summarises the firms' profits for different market structures.

The (1, 1) game, where each firm has a current bid of 1 unit, has the payoff structure illustrated in Table 10. This game has a mixed strategy equilibrium (provided $\frac{17}{9} \geq P_a \geq \left(\frac{26}{27}\right)^2$) with probabilities of dropping:

$$\pi_i^*((1,1),P_a) = \frac{P_a - \frac{676}{729}}{\frac{1}{2}P_a + \frac{25}{1458}}.$$

For $P_a > \frac{17}{9}$, the firms both exit the auction, while for $P_a < \left(\frac{26}{27}\right)^2$, neither will drop, and the auction will clear with a duopoly in the downstream market.

An analogous situation to Fig. 1 prevails, where for low prices, the auction will end instantly, whereas for higher prices, the firms will wish to delay and allow the mixed equilibrium to take place. Because the firms are symmetric, both agree about the optimal price at which they should drop to (1, 1). Hence, setting initial caps of (2, 1) or (2, 2) would result in the same equilibrium. The bottom graph of Fig. 2 shows the outcomes of this auction, as a function of reserve price.

A2. Case 3: only one firm is low cost

In this case, the firms are asymmetric: without loss of generality, firm 1 is low cost ($c_1 = 1$), while firm 2 is high cost ($c_2 = 10/9$). The downstream market payoffs are provided in the third block (columns 6 and 7) in Table 1. The firms have different payoffs in the $\mathbf{B} = (1,1)$ game; see Table 11. Consequently, the two firms will *not* be indifferent between $\mathbf{B} = (2,1)$ and $\mathbf{B} = (1,2)$.

A2.1. The (1,1) game

In the (1, 1) game, each firm has a current bid of 1 unit. This game has different equilibria depending on P_a:

1. If $P_a < \left(\frac{17}{18}\right)^2$, then the pure strategy Nash equilibrium is: *(Not Drop, Not Drop)*, which results in a duopoly.
2. If $\left(\frac{17}{18}\right)^2 < P_a < \frac{19}{18}$, then the pure strategy Nash equilibrium is: *(Not Drop, Drop)*, which results in a monopoly (firm 1) with 1 unit.
3. If $\frac{19}{8} < P_a < \frac{17}{9}$, then the firms no longer have dominant strategies, so there is a mixed strategy equilibrium.
4. If $\frac{17}{9} < P_a < 2$, then the pure strategy Nash equilibrium is: *(Not Drop, Drop)*, which results in a monopoly (firm 1) with 1 unit.
5. If $P_a > 2$, then the pure strategy Nash equilibrium is: *(Drop, Drop)*, which results in no units being sold, as the auction price is too high.

We display the payoffs to the two firms in Fig. 3.

A2.2. The (1,2) game

In the (1,2) game, firm 1 has a current bid of 1 unit, while firm 2 bids 2. Therefore, firm 2 controls when the auction will proceed to $\mathbf{B} = (1, 1)$ as discussed in Section 3.2 (see Fig. 3 for intervals discussed):

1. If $P_a \in$ VII, then firm 2 chooses *Drop* and immediately ends the auction. This results in a duopoly.
2. If $P_a \in$ X ($2 > P_a > 17/19$), then firm 2 will drop, and subsequently drop again in the (1,1) game. Firm 1 will become a monopolist.
3. If $P_a \in$ IX, then firm 2 will immediately drop, the two firms will mix, and a duopoly or monopoly could result.
4. If $P_a \in$ VIII, then firm 2 has an incentive to wait, and drop only when P_a reaches sup(VIII).
5. If $P_a > 2$, firm 2 exits the auction, along with firm 1.

A2.3. The (2,1) game

In the (2,1) game, firm 1 has a current bid of 2 units, while firm 2 bids 1. Firm 1 has the following equilibrium strategy (see Fig. 3):

1. If $P_a \in$ I ($P_a < (17/18)^2$), firm 1 can wait until $P_a = (17/18)^2$, and then lower demand, with the resulting game giving it a monopoly with 1 unit. In doing so, firm 2 will win nothing. Hence at $P_a < (17/18)^2$, firm 2 is indifferent between dropping and not dropping. For sufficiently small price increments, this is equivalent to firm 2 exiting the auction immediately, and firm 1 has the option to keep both units, which it exercises if $P_a < 1/4$.
2. If $P_a \in$ II ($(17/18)^2 < P_a < 19/18$), then firm 1 will drop, followed by firm 2 dropping in the (1,1) game. Finally, firm 1 becomes a 1-unit monopolist.

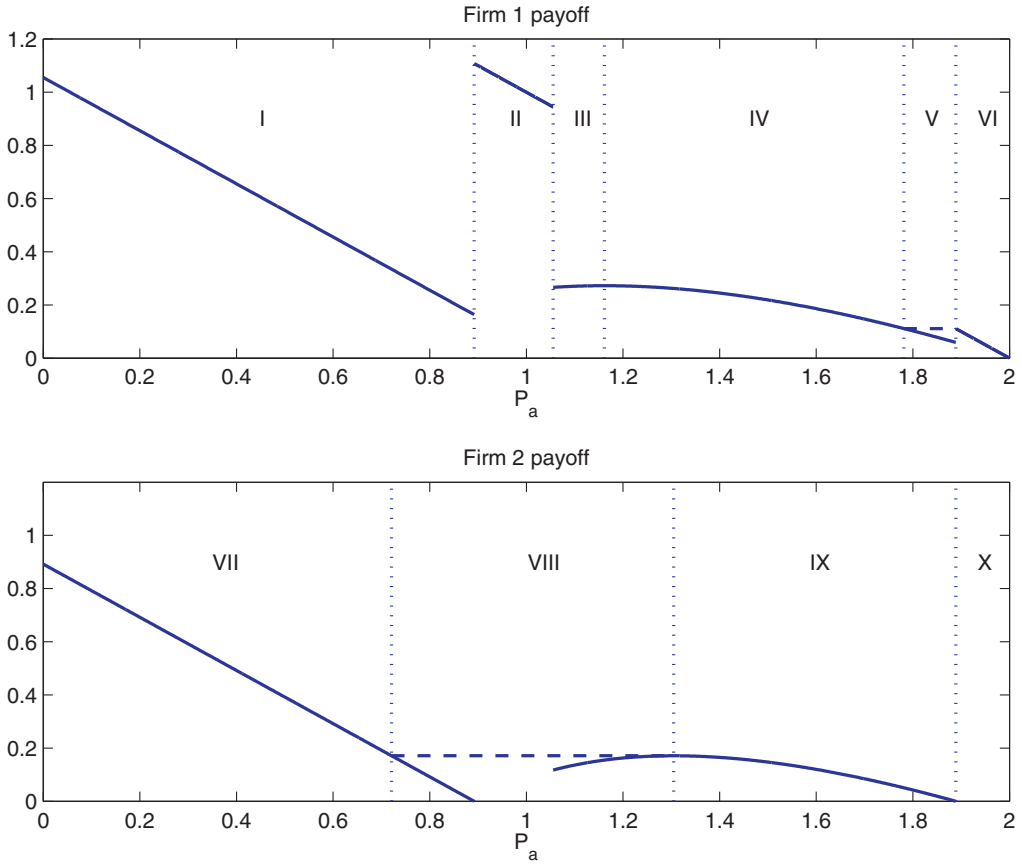


Fig. 3. Firm expected payoffs for the two firms if the bid vector is one unit each and the price is P_a (i.e., $V_i^*((1,1),P_a)$). Firm 1 is low cost, while firm 2 is high cost ($c_1 = 1$, $c_2 = \frac{10}{9}$). The top graph shows firm 1's expected payoff, while the bottom graph shows firm 2's expected payoff. Dotted lines divide the price level into intervals.

3. If $P_a \in$ III, firm 1 waits until $P_a = \sup(\text{III})$.
4. If $P_a \in$ IV, firm 1 immediately drops demand, which then leads to a mixed strategy equilibrium in the (1,1) game.
5. If $P_a \in$ V, firm 1 waits until $P_a = 17/9$, at which price both firms drop demand by 1 unit, resulting in a monopoly.
6. If $P_a \in$ VI ($17/9 < P_a < 2$), firm 1 ends the auction immediately, resulting in a monopoly with 1 unit.
7. If $P_a > 2$, firm 1 exits the auction (along with firm 2).

A2.4. The (2,2) game

We can finally consider what happens if $\mathbf{B} = (2,2)$.

1. If $P_a \in$ VI then firm 2 knows that it will win no units. Hence it will exit the auction, and firm 1 will win 1 unit.

2. If $P_a \in$ V, firm 1 will delay ending the auction until $P_a = 17/9$, whereupon it will buy one unit, and firm 2 will exit the auction.
3. If $P_a \in \text{IV} \cap \text{IX}$, then both firms agree that the auction should end now, and the firms will mix in the (1,1) game.
4. If $P_a \in \text{IV} \cap \text{VIII}$, then firm 1 favours dropping, but firm 2 favours waiting. The auction will thus progress to $P_a = \sup(\text{VIII})$, and then end.
5. If $P_a \in$ III, then both firms initially favour waiting, and the price will rise, leading to the previous case, and the auction will end at $P_a = \sup(\text{VIII})$.
6. If $P_a \in$ II, then firm 1 would like to end the auction, but firm 2 will delay, again leading to the auction closing at $P_a = \sup(\text{VIII})$.
7. If $P_a \in \text{VIII} \cap \text{I}$, then firm 2 still favours delay, and the auction will clear at $P_a = \sup(\text{VIII})$.
8. If $P_a \in$ VII, if firm 1 does not drop, firm 2 will delay and attain its local maximum payoff ($P_a = \sup(\text{VIII})$). Firm 1 would then receive a lower payoff than it achieves at any price in region VII. Further, firm 2's best response to firm 1 dropping demand is to end the auction. Hence firm 1 will drop immediately to terminate the auction when $P_a \in$ VII (resulting in a duopoly).

A3. 2 units, 2 firms, and an uncertain regulator

From an auction design perspective, we can examine the expected payoffs to starting the auction at $\overline{\mathbf{B}} = (2,2)$, $\overline{\mathbf{B}} = (2,1)$, $\overline{\mathbf{B}} = (1,2)$, $\overline{\mathbf{B}} = (1,1)$, $\overline{\mathbf{B}} = (1,0)$, or $\overline{\mathbf{B}} = (0,1)$, as a function of reserve price. Fig. 4 plots payoffs from starting the asymmetric firm auction at a given price and cap combination.

We can combine Figs. 2 and 4 to consider the optimal behaviour of a regulator who is uncertain of the type of firms he/she is presiding over.[23] We assume the firms are independently distributed, and each has 50% probability of being high or low cost.

Capping at (2,1) or (1,2) will not affect the regulator's outcome if the firms are symmetric (see Section 3 and Appendix A.1) since both firms agree about the price at which they would prefer to play the (1,1) game. However, if the firms are asymmetric, imposing a (2,1) or (1,2) cap may achieve either the solid line payoff in Fig. 4 or the dashed line. Both outcomes will be equally likely, from the regulator's perspective. The regulator's payoff to using either (2,1) or (1,2) caps is thus an equally weighted combination of the case 1, $\overline{\mathbf{B}} = (2,1)$ payoff; the case 2, $\overline{\mathbf{B}} = (2,1)$ payoff; the case 3, $\overline{\mathbf{B}} = (2,1)$ payoff; and the case 3, $\overline{\mathbf{B}} = (1,2)$ payoff. A similar analysis applies to using (1,0) or (0,1) caps.

We can therefore create a new set of graphs for an uncertain regulator, showing his/her payoff from either running an evenly capped auction or favouring (blindly) one of the firms. The optimal cap and reserve price strategy will be the combination that maximises the regulator's objective function. Fig. 5 shows three regulator expected payoffs (expected total surplus, expected revenue, and an average of the two).

[23] Behaviour for the (1,1), (1,0), and (0,1) auctions is easily calculated using Tables 1, 2, 10, and 11.

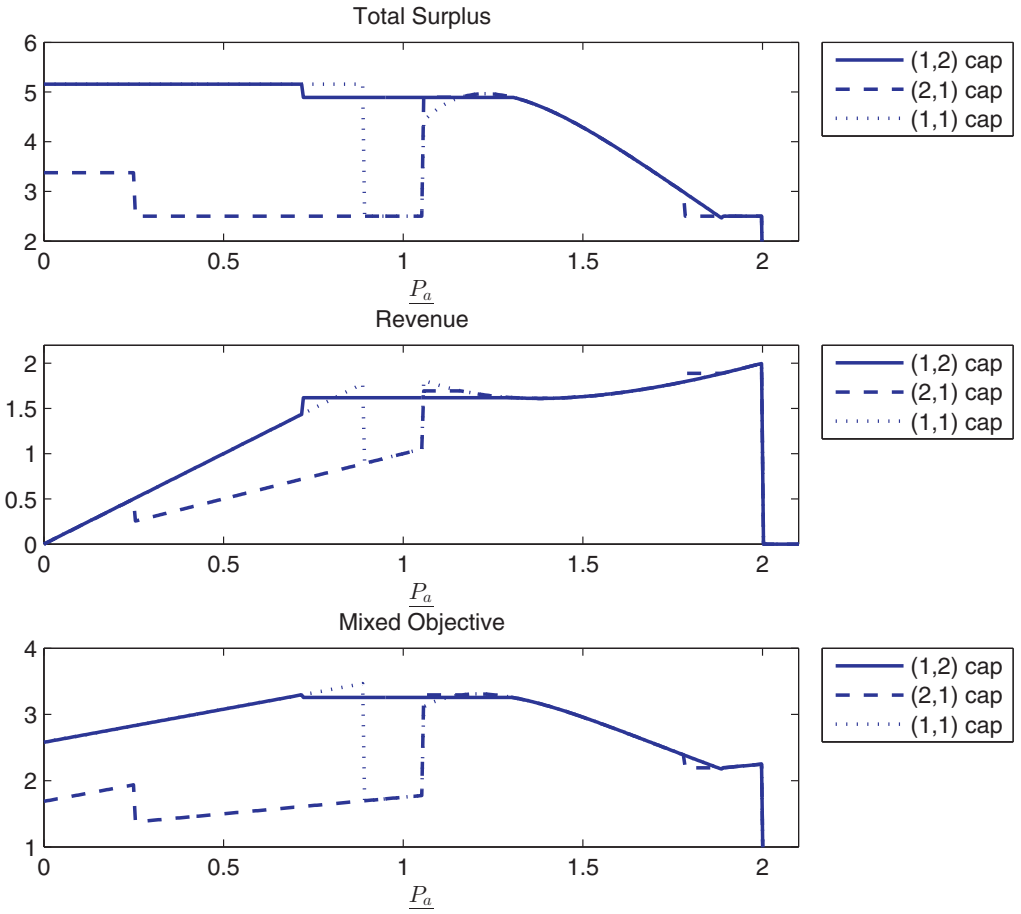


Fig. 4. Regulator expected payoffs (as a function of reserve price and capping), when firm 1 is low cost, and firm 2 is high cost ($c_1 = 1$, $c_2 = 10/9$). The three graphs show different regulator objectives, while the lines denote cap policies. The horizontal axes measure reserve price. The (2,2) cap outcome (not shown) is coincident with the (1,2) outcome for all $\underline{P_a} < 1.5$. For $\underline{P_a} \geq 1.5$ the (2,2) cap outcome is coincident with the (2,1) cap. For $\underline{P_a} \geq 1.5$, the (1,1) cap is coincident with the (1,2) cap. The (1,0) and (0,1) cap cases are not shown. The (1,0) cap yields total surplus $\frac{5}{2}$ if $\underline{P_a} < 2$, 0 otherwise. The (0,1) cap yields total surplus $\frac{43}{18}$ if $\underline{P_a} < \frac{17}{9}$, 0 otherwise. Both (1,0) and (0,1) caps give revenue $\underline{P_a}$ provided total surplus is positive.

Generally (but not always) for a given reserve price, the regulator is better to run an uncapped or (1,1) capped auction. For reserve prices where case 3 (the asymmetric firm case) could result in a low cost firm becoming a monopolist ($\left(\frac{17}{18}\right)^2 < \underline{P_a} < \frac{19}{18}$), an uncapped auction may be preferred, to allow the high cost firm to force the price higher. Overall, the optimal behaviour for a total surplus maximising regulator is to run a (2,2) capped auction with $\underline{P_a} \in$ VII, or a (1,1) capped auction with $\underline{P_a} \in$ I. A revenue maximising regulator would opt for a high auction price, such that even if both firms are high cost, there will still be units sold ($\underline{P_a}$ just below $\frac{17}{9}$)). The revenue maximising regulator is (at this reserve price) indifferent to the cap structure. In contrast, a regulator

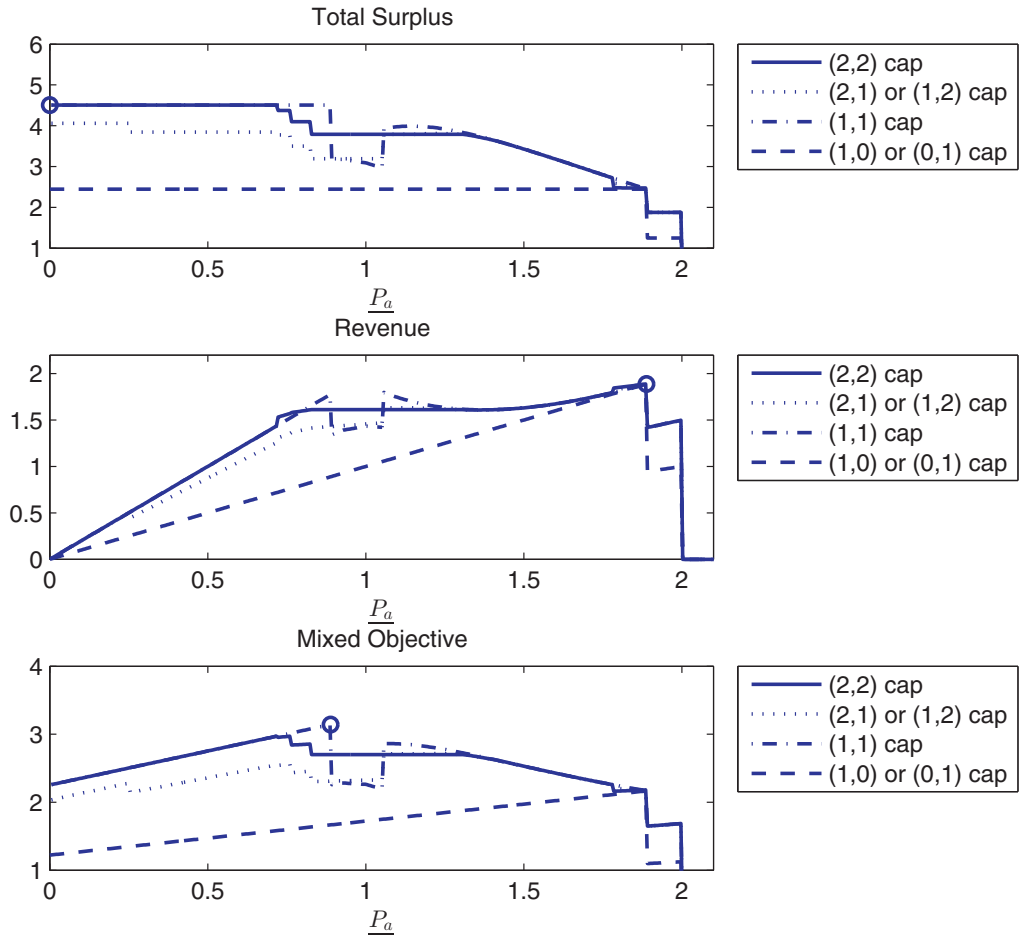


Fig. 5. Regulator expected payoffs where costs are uncertain, as a function of reserve price and cap structure. Each firm has independent costs, with probability 1/2 of cost 1 and probability 1/2 of cost 10/9. Circle denotes maximum point. The three graphs show different regulator objectives, while the lines denote cap policies. The horizontal axes measure reserve price.

with a mixed objective will run a (1,1) capped auction where the reserve price is chosen such that if the firms are asymmetric, there is no danger of the low cost firm becoming a monopolist ($\underline{P_a}$ is infinitesimally below $\left(\frac{17}{18}\right)^2$).

Appendix B. Solving the auction

B1. Solving the dynamic programming problem

To calculate V_i^* and π_i^*, we proceed by working backwards, from the maximal price of the auction to the reserve price. For each price level, we work through the different combinations of $\mathbf{B}$ sequentially, beginning with cases where $\sum_{i=1}^{M} B_i = 0$, before proceeding to cases where $\sum_{i=1}^{M} B_i = 1$, then $\sum_{i=1}^{M} B_i = 2$, etc. First, we define the *continuation*

value ($\widehat{V}_i(B, P_a)$), which is attained if no firm reduces demand at the current price:

$$\widehat{V}_i(\mathbf{B}, P_a) = \begin{cases} V_i^*(\mathbf{B}, P_a + \Delta P, 0) & \text{if } \sum_{i=1}^{M} B_i > N \\ \overline{V}_i(\mathbf{B}, P_a) & \text{if } \sum_{i=1}^{M} B_i \leq N. \end{cases} \tag{B.1}$$

In Eq. (B.1), the continuation value equals the terminal value (see Eq. (7)), if there is no excess demand at the current price (in which case the auction ends). Otherwise, the auction continues with a higher price. Before we define the value function for firm i during the auction, it is important to introduce one more notation: the change in firm i's expected payoff due to firm j dropping its demand by one unit $\Delta V_{ij}(\mathbf{B},\ P_a)$ is:

$$\Delta V_{ij}(B_j, \mathbf{B}_{-j}, P_a) = V_i^*(B_j - 1, \mathbf{B}_{-j}, P_a, j) - \widehat{V}_i(\mathbf{B}, P_a) \tag{B.2}$$

where $\Delta V_{ij}(\ \cdot\ )$ applies for cases where $B_j \geq 1$.[24]

$-\Delta V_{ii}$ can be viewed as a *net local* marginal valuation of a unit. It is *net* in the sense that it is a valuation in excess of price. It is *local* in that it is the *expected* effect after taking into account all other future bidding behaviour by firm i and other firms. In some cases $-\Delta V_{ii}$ is negative, where dropping a unit may be in firm i's interest, to avoid the auction price rising (and affecting all of firm i's winnings) or where the auction may close but firm i may pay more for a unit than it will earn from that unit in the downstream market.

Note that any reduction in demand results in the auction moving to a new level of demand, and that the price remains at the same level, which allows further bid reductions to take place.

Suppressing arguments, suppose that each firm chooses to drop its demand with probability π_i, with the vector of probabilities given by $\boldsymbol{\pi}$. Then firm i's expected payoff is given by:

$$V_i(\mathbf{B}, P_a, \boldsymbol{\pi}, 0) = \widehat{V}_i(\mathbf{B}, P_a) + \sum_{\substack{\delta_1,\ldots,\delta_M \in \{0,1\}^M \\ \delta_1+\cdots+\delta_M \neq 0 \\ \delta_1 \leq B_1,\ldots,\delta_M \leq B_M}} \left\{ \left[\prod_{k=1}^{M} \pi_k^{\delta_k} (1-\pi_k)^{1-\delta_k} \right] \left[\frac{\sum_{j=1}^{M} \Delta V_{ij}(\mathbf{B}, P_a)\ \delta_j}{\sum_{j'=1}^{M} \delta_{j'}} \right] \right\} \tag{B.3}$$

where δ_i is a binary variable that represents whether firm i drops its demand or not. The case $\phi = 0$ implies that no firm has just lowered demand, so each firm that aims to drop demand is equally likely to do so successfully. In Eq. (B.3), the first term is the case where no firm drops demand, whereas the second term (starting with the summation) is the case where at least one firm tries to drop demand. In particular, the first component in the summation is the probability of seeing a given combination of firms *trying* to drop their demand, while the final fraction weights the change in firm i's expected payoff from each demand reduction by the probability that each succeeds, conditional upon trying.

[24] ΔV_{ij} is not a function of ϕ, since both terms on the right-hand side do not depend on which firms have previously dropped.

If firm m has just dropped its demand by 1 (and therefore can engage in sequential dropping), then firm i's current valuation is:

$$V_i(\mathbf{B}, P_a, \boldsymbol{\pi}, m) = \widehat{V}_i(\mathbf{B}, P_a) + \pi_m \, \Delta V_{im}(\mathbf{B}, P_a) + (1 - \pi_m) \left(V_i^*(\mathbf{B}, P_a, 0) - \widehat{V}_i(\mathbf{B}, P_a) \right); \forall m = 1, \ldots, M. \tag{B.4}$$

Given the current valuation, if firm i follows a mixed strategy, then it is indifferent between lowering its own demand and maintaining it at the current level:

$$V_i(\mathbf{B}, P_a, \pi_i = 1, \boldsymbol{\pi}_{-i}, \phi) = V_i(\mathbf{B}, P_a, \pi_i = 0, \boldsymbol{\pi}_{-i}, \phi). \tag{B.5}$$

Next, we define the mixed-strategy Nash equilibrium using the system of equations (B.5).

Definition B.1. Nash Equilibrium in mixed strategies for the downstream market and the auction with discrete bidding is such that, given ($\mathbf{B}$, P_a, ϕ), for each firm i, there exists a probability $\pi_i^* = \pi_i^*(\mathbf{B}, P_a, \phi)$ that maximises firm i's expected payoff $\forall i = 1, \ldots, M$:

$$V_i(\mathbf{B}, P_a, \pi_i^*, \boldsymbol{\pi}_{-i}^*, \phi) \geq V_i(\mathbf{B}, P_a, \rho_i, \boldsymbol{\pi}_{-i}^*, \phi); \forall \rho_i \in [0, 1]. \tag{B.6}$$

The value function for each firm is given by

$$V_i^*(\mathbf{B}, P_a, \phi) = V_i(\mathbf{B}, P_a, \boldsymbol{\pi}^*, \phi).$$

B2. Solving the finite state game

Solving for $\boldsymbol{\pi}^*$ implies solving a system of equations, each of which is represented by (B.6) with the inequality binding, for each of the firms who is following a mixed strategy. For the dynamic programme to be solvable, we must have a unique value function for each firm in each state. In many cases, however, there are multiple equilibria for (B.6). In these cases, where possible, we choose a mixed strategy equilibrium, since it will be symmetric when (B.6) is symmetric. Failing this, we choose pure equilibria who have higher risk measures (argued by Carlsson and van Damme, 1993 to represent more likely equilibria). In the event of a tie in risk measures, we assume that each pure strategy equilibrium is equally likely to occur.[25] Failing pure strategy equilibria, we will accept semi-mixed equilibria, where each is assumed equally likely to occur.

At each state, described by an auction price, a vector of current bids, and the firm who last lowered demand, each firm decides whether to drop its demand by one unit or not, given the others' actions. Let F denote the set of players (firms) and each player has only two actions (*Drop* and *Not Drop*). We describe the pseudo-algorithm below to solve the finite state games:

Step 1 Check for active players: let F_0 denote the set of players with zero current bids. The set of active players is: $F_1 = F \backslash F_0$.

[25] See our example in Section 3 and Appendix A for some illustrative cases.

Step 2 For each player in F_1, check if there is a dominant strategy, or if the firm is indifferent between actions. Let $\bar{F}$ denote the set of such players with a dominant strategy or who are indifferent. The set of players who could play a mixed strategy is $F_2 = F_1 \backslash \bar{F}$. Firms who are indifferent are assumed to drop with probability 0.5.

Step 3 For the players in F_2, solve for the mixed-strategy Nash equilibrium (MSNE) in two ways:

1. Root-finding method: compute the roots of the system of non-linear equations (B.5), and check if the root (i.e., probability of dropping $\{\pi_i\}_{i \in F_2}$) satisfies the following condition: $\pi_i \in (0, 1), \forall i \in F_2$.[26]
2. Non-linear optimisation method: if the root-finding method does not reveal a solution that satisfies the condition, then solve the nonlinear Nash equilibrium constrained optimisation problem, with a predefined set of initial conditions.

Step 4 If the two methods in Step 3 do not yield an MSNE, look for a pure-strategy Nash equilibrium (PSNE).

1. If there is a single PSNE, then accept it as the solution to the game;
2. If there are multiple PSNEs, then rank them according to their risk measure (Carlsson and van Damme, 1993): $R_j = \prod_{i=1}^{F_2} (P_{s_{ij}} - P_{s'_{ij}})$ where s_{ij} is the strategy played in PSNE j and s'_{ij} is the one-step deviation from s_{ij} for player i. The risk measure R_j for PSNE j is the product of the difference between equilibrium payoff ($P_{s_{ij}}$) and the corresponding one-step deviations ($P_{s'_{ij}}$). The PSNE with the highest risk measure is the solution to the game. If multiple PSNEs have identical risk measures, we assume they occur with equal probability.

Step 5 If no MSNE or PSNE are found in Steps 3 and 4, then look for a semi-mixed solution where some players are constrained to play pure strategies and others are allowed to play mixed strategies. For each combination, we check whether the outcome is a Nash equilibrium, and assume that each occurs with equal probability.

B3. Sequential dropping

In most auctions, firms may lower their bid by multiple units. We model this by allowing *sequential dropping*: if a firm drops demand by one unit, then that firm has the option to drop its demand further before any other firms react. For example, a firm dropping its demand by five units would be modelled as five sequential drops of one unit, with no opportunity for others to intervene.

[26] We solve this system of equations using Newton's method. It is helpful to note that

$$\frac{\partial V_i(\mathbf{B}, P_a, \boldsymbol{\pi}, 0)}{\partial \pi_n} = \sum_{\substack{\delta_1,\ldots,\delta_M \in \{0,1\}^M \\ \delta_1+\ldots+\delta_M \neq 0 \\ \delta_1 \leq B_1,\ldots,\delta_M \leq B_M}} (2\delta_n - 1) \left[\prod_{k \neq n} \pi_k^{\delta_k} (1 - \pi_k)^{1-\delta_k} \right] \left[\frac{\sum_{j=1}^{M} \Delta V_{ij}(\mathbf{B}, P_a)\ \delta_j}{\sum_{j'=1}^{M} \delta'_j} \right].$$

B4. Incorporating caps and reserve price

Given that our solution technique (in Appendix B.1) works recursively backwards from high prices and low demand to low prices and high demand, it is easy to modify the auction to incorporate caps (see Section 2.2). For each combination of starting bids and reserve price, it is possible to evaluate the distribution of auction clearing bids. These allow us to calculate the distribution of total surplus and revenue if demand and supply parameters are known with certainty.

References

Binmore, K., Klemperer, P., 2002. The biggest auction ever: the sale of the British 3G telecom licences. Economic Journal 112, C74–C96.
Blume, A., Heidhues, P., Lafky, J., Münster, J., 2009. All equilibria of the multi-unit Vickrey auction. Games and Economic Behavior 66, 729–741.
Bush, C.A., 2010. Increasing diversity in telecommunications ownership and increasing efficiency in spectrum auctions by breaking the link between capital market discrimination and FCC spectrum auction outcomes. Review of Black Political Economy 37, 131–152.
Carlsson, H., van Damme, E., 1993. Global games and equilibrium selection. Econometrica 61, 989–1018.
Cramton, P., 1997. The FCC spectrum auctions: an early assessment. Journal of Economics and Management Strategy 6, 431–495.
Cramton, P., 1998. The efficiency of the FCC spectrum auctions. Journal of Law and Economics 41, 727–736.
Cramton, P., 2002. Spectrum auctions. In: Cave, M., Majumdar, S., Vogelsang, I. (Eds.), Handbook of Telecommunications Economics, Chapter 14. Elsevier Science B.V., Amsterdam, pp. 605–639.
Cramton, P., 2013. Spectrum auction design. Review of Industrial Organization 42, 161–190.
Cramton, P., Kwerel, E., Rosston, G., Skrzypacz, A., 2011. Using spectrum auctions to enhance competition in wireless services. Journal of Law and Economics 54, S167–S188.
van Damme, E., 2002. The European UMTS-auctions. European Economic Review 46, 846–869.
Dana, J.D., Spier, K.E., 1994. Designing a private industry: government auctions with endogenous market structure. Journal of Public Economics 53, 127–147.
Eső, P., Nocke, V., White, L., 2010. Competition for scarce resources. RAND Journal of Economics 41, 524–548.
Grimm, V., Riedel, F., Wolfstetter, E., 2003. Low price equilibrium in multi-unit auctions: the GSM spectrum auction in Germany. International Journal of Industrial Organization 21, 1557–1569.
Hoppe, H.C., Jehiel, P., Moldovanu, B., 2006. License auctions and market structure. Journal of Economics and Management Strategy 15, 371–396.
Hortaçsu, A., McAdams, D., 2010. Mechanism choice and strategic bidding in divisible good auctions: an empirical analysis of the Turkish treasury auction market. Journal of Political Economy 118, 833–865.
Janssen, M., 2006. Auctions as coordination devices. European Economic Review 50, 517–532.
Janssen, M., Karamychev, V., 2007. Selection effects in auctions for monopoly rights. Journal of Economic Theory 134, 576–582.
Jehiel, P., Moldovanu, B., 2003. An economic perspective on auctions. Economic Policy 18, 269–308.
Kastl, J., 2011. Discrete bids and empirical inference in divisible good auctions. Review of Economic Studies 78, 974–1014.
Katzman, B., 1999. A two stage sequential auction with multi-unit demands. Journal of Economic Theory 86, 77–99.
Klemperer, P., 2005. How (not) to run auctions: the European 3G telecom auctions. European Economic Review 46, 229–248.
Kwerel, E.R., Rosston, G.L., 2000. An insiders' view of FCC spectrum auctions. Journal of Regulatory Economics 17, 253–289.
Laye, J., Laye, M., 2008. Uniqueness and characterization of capacity constrained Cournot–Nash equilibrium. Operations Research Letters 36, 168–172.

Mishra, D., Parkes, D., 2009. Multi-item Vickrey–Dutch auctions. Games and Economic Behavior 66, 326–347.
Monderer, D., Shapley, L., 1996. Potential games. Games and Economic Behavior 14, 124–143.
Offerman, T., Potters, J., 2000. Does Auctioning of Entry Licenses Affect Consumer Prices? An Experimental Study. Working Paper.
Riedel, F., Wolfstetter, E., 2006. Immediate demand reduction in simultaneous ascending-bid auctions: a uniqueness result. Economic Theory 29, 721–726.